CGLMP BOUNDS, MEASUREMENT RIGIDITY, AND STATISTICAL STRENGTH IN TWO-SETTING BELL TESTS

Abstract. We determine the sharp CGLMP bound for projective measurements on maximally entangled states in the bipartite Bell scenario with two settings and d outcomes; these questions form Problem 27 of the IQOQI list of open problems in quantum information. Equality holds precisely when the local dimension is a multiple of d and the projective measurements are unitarily equivalent to Fourier-phase measurements with the corresponding multiplicity. Quantitative estimates control the measurement distance by the Bell-value deficit and give an arithmetic dimension obstruction. The proof combines a matrix Hardy inequality with a positive law on partitions and polynomial positivity bounds. We then compare CGLMP optimization with noise robustness and Kullback–Leibler statistical strength. Fourier-phase projective measurements optimize uniform-noise tolerance for the CGLMP witness, whereas two-setting tensor measurements achieve greater global white-noise resistance and statistical strength. For every input distribution, we determine the dimension-unrestricted statistical-strength supremum and give a sequence approaching it. A constructive local-component bound places every finite-outcome quantum behaviour strictly below this supremum for full-support inputs.

1. Introduction

Bell experiments with two measurement choices and many outcomes offer a direct way to study nonlocality in higher-dimensional quantum systems. Increasing the number of outcomes makes the internal structure of a qudit accessible while keeping the input alphabet small. This combination is also useful for device-independent certification, where a small number of measurement choices can certify a large local Hilbert space [1, 2]. It motivates the search for Bell inequalities whose optimal measurements admit both an experimentally natural implementation and a mathematical characterization.

The CGLMP inequalities of Collins, Gisin, Linden, Massar, and Popescu [3] provide a distinguished family in this setting. Their local bound follows from a modular constraint around the four measurement contexts, and each inequality defines a facet of the local polytope [4]. The associated Fourier-phase measurements combine a discrete Fourier transformation with diagonal unitaries. Durt, Kaszlikowski, and Żukowski [5] conjectured that these measurements are optimal on the maximally entangled state. Zohren and Gill [6] subsequently examined the same measurement family in their analysis of large-dimensional CGLMP violations. Beyond the optimal value, a natural rigidity question asks whether every maximizing projective strategy has this form.

The fixed-state problem has a useful tracial structure: correlations on a maximally entangled state are normalized traces of products of measurement operators. Lang, Vértesi, and Navascués [7] exploited this structure in semidefinite relaxations and obtained sharp numerical evidence for the three-outcome value. Optimization over states leads to a different problem. Acín, Durt, Gisin, and Latorre [8] showed that nonmaximally entangled states can give larger CGLMP violations. The hierarchy of Navascués, Pironio, and Acín [9], together with symmetry-adapted polynomial methods [10], provides arbitrary-state bounds, including tight analytic certificates at $d = 3, 4$. Numerical black-box certification of the qutrit optimum was developed by Yang et al. [11]. Related rigidity results concern mutually unbiased bases in Bell scenarios, with three settings in the qutrit case [12], and in prepare-and-measure scenarios [13].

A second issue is the meaning of an optimal Bell experiment. A large value of a chosen functional determines the amount of uniform noise that this particular witness tolerates. Global white-noise resistance instead depends on the intersection of the noise line with the whole local polytope. Statistical strength is a third criterion: the quantum-to-local Kullback–Leibler divergence quantifies the expected evidence against local realism per complete trial [14, 15]. It depends on the entire behaviour and on the distribution of the inputs. Acín, Gill, and Gisin [15] compared these choices in the design of Bell tests, while Baek, Ryu, and Lee [16] studied distinctions between robustness measures. These considerations make it important to keep the witness, the noise model, and the statistical objective separate.

The questions studied in this paper were codified by Gill as Problem 27 of the IQOQI list of open problems in quantum information [17]. We reproduce its statement, with the citation markers of the original resolved to the present bibliography.

Problem 27 (IQOQI Open Quantum Problems): The power of CGLMP inequalities

In the setting of Problem 26, consider especially the case $(N, M, K) = (2, 2, d)$.

Problem 27. A. Show that every face of the local polytope C, which is not already contained in a face of the no-signalling polytope P is of CGLMP type, i.e., an inequality of the form first written out in [3], but possibly lifted from lower dimensions by fusing together some outcomes.

Problem 27. B. Numerically, the observables maximally violating the CGLMP inequality on a maximally entangled state are of a very specific form [5], involving measurements in computational basis, transformed by only discrete Fourier transformation and diagonal unitaries [3]. Show that this is necessarily the case. Show also that these measurements realize the highest resistance of violation to noise, and the best discrimination against classical realism in the sense of Kullback–Leibler divergence [14].

Here N, M and K denote the numbers of parties, settings per party and outcomes per measurement, and Problem 26 of the same list concerns Bell inequalities in this general setting. The answers are as follows. Part A fails, as Bancal, Gisin, and Pironio already observed in answering Gill's question in the negative [18]: their four-outcome facet, recalled in Appendix A (Proposition A.1) with a self-contained certificate, lies in no face of the no-signalling polytope and is neither a CGLMP inequality nor a lifting of one. In Part B, the first assertion holds for projective measurements: Theorem 1.1 shows that on a maximally entangled state the CGLMP value is maximized only by Fourier-phase measurements, in every outcome count, and Theorem 1.2 makes this quantitative. The two optimality assertions of Part B hold only in a restricted sense: Theorem 1.3 shows that Fourier-phase measurements optimize uniform-noise tolerance for the CGLMP witness itself but not global white-noise resistance, and Theorem 1.4 shows that they do not optimize Kullback–Leibler statistical strength, whose dimension-unrestricted supremum we determine for every input distribution. At two outcomes, Fine's characterization gives the full local description [19]; higher-outcome facet enumeration already exhibits a richer geometry [20, 18, 21, 22].

Main results. We normalize the CGLMP functional I_d to have local bound 2. Write $\Phi_n = n^{-1/2}\sum_{j=0}^{n-1}|jj\rangle$. The standard Fourier-phase projectors Π_a^t are defined in (2.2). For the fixed-state statements, we use the tracial ordering

$$(P^0, P^1, P^2, P^3) = (A_0^T, B_0, A_1^T, B_1).$$

Theorem 1.1 (Projective value and rigidity)**.** *Let $d \geq 2$, $n \geq 1$, and let the four measurements on Φ_n be d-outcome PVMs. Then*

$$I_d \leq I_d^{\mathrm{F}} := \frac{4}{d(d-1)} \sum_{h=1}^{d-1} (d-h) \sec\left(\frac{\pi h}{2d}\right). \tag{1.1}$$

Equality holds if and only if $n = dk$ and, after a common unitary conjugation in the tracial representation, the four PVMs are $\Pi_a^t \otimes I_k$. In particular, all projective maximizers on Φ_d are rank-one Fourier-phase measurements.

Sarkar et al. [2] obtained a device-independent self-test of the same measurements for the inequalities of Salavrakos et al. [1]. Theorem 1.1 is a rigidity statement for projective measurements on the fixed maximally entangled state and for the CGLMP functional itself.

The distance $\mathrm{Err}_{d,n}$ and constants $C_d^{\mathrm{st}}, R_d^{\mathrm{st}}, K_d$ are defined in (2.3) and (2.4).

Theorem 1.2 (Quantitative rigidity and arithmetic obstruction)**.** *Let $d \geq 2$, $n \geq 1$, and let P be the tracial PVMs of d-outcome projective measurements on Φ_n. Put $\Delta = I_d^{\mathrm{F}} - I_d$. If $d \mid n$, then*

$$\mathrm{Err}_{d,n}(P) \leq \min\{8, K_d \Delta\}, \qquad K_d \leq 786432\, d^6. \tag{1.2}$$

For arbitrary n, with $s = n \bmod d$,

$$\Delta \geq \frac{s(d-s)}{3n\, d^4(d+1) C_d^{\mathrm{st}}}. \tag{1.3}$$

If $\mathcal{I}_{d,n}$ denotes the projective maximum in that dimension, then for $s > 0$,

$$\frac{s(d-s)}{3n\, d^4(d+1) C_d^{\mathrm{st}}} \leq I_d^{\mathrm{F}} - \mathcal{I}_{d,n} \leq \frac{s}{n}(I_d^{\mathrm{F}} - 2). \tag{1.4}$$

The arithmetic obstruction distinguishes the ambient dimension from the number of outcomes: equality requires $d \mid n$, while (1.4) controls the optimal deficit when d does not divide n.

Write Q_d for the Fourier-phase behaviour in (6.1) and T_m for the common-input tensor CHSH behaviour in (6.5). The quantity $v_{\mathrm{crit}}(q)$ denotes the critical visibility of q under uniform noise (Section 2).

Theorem 1.3 (Witness tolerance and global noise resistance)**.** *For d-outcome projective measurements on Φ_n producing q with $I_d(q) > 2$,*

$$v_{\mathrm{CGLMP}}(q) = \frac{2}{I_d(q)} \geq \frac{2}{I_d^{\mathrm{F}}}. \tag{1.5}$$

Equality holds precisely for the standard amplifications in Theorem 1.1; in particular, the minimum is attained whenever $d \mid n$. For every integer $m \geq 20000$, the two rank-one measurements per party defining T_m on Φ_{2^m} satisfy

$$v_{\mathrm{crit}}(T_m) \leq \frac{17}{29} < \frac{3}{5} < v_{\mathrm{crit}}(Q_{2^m}). \tag{1.6}$$

The comparison keeps the state, local dimension, number of settings, and measurement ranks fixed. Common-input tensor measurements also appear in the certification of tensor products with an unchanged number of measurement choices [23].

For an input law μ, let $\mathcal{S}_\mu$ be the statistical strength in (2.5), and let $\Xi(\mu)$ be the four-variable minimum in (2.6).

Theorem 1.4 (Dimension-unrestricted statistical strength)**.** *For every fixed input distribution independent of the hidden variable,*

$$\sup_{\substack{d \geq 2 \\ \text{two rank-one PVMs per party on } \Phi_d}} \mathcal{S}_\mu(q) = \Xi(\mu). \tag{1.7}$$

The same supremum holds over arbitrary finite-dimensional quantum states and finite-outcome POVMs. For full-support μ, every finite-outcome quantum behaviour has statistical strength strictly smaller than $\Xi(\mu)$.

For uniform inputs, the ceiling is $\log_2(4/3)$, the relative entropy of the PR box computed by Grudka et al. [24]. The finite-outcome gap follows from $\mathrm{LF}(q) \geq d - \sqrt{d^2-1} > 1/(2d)$, a quantitative local-component bound related to the two-input results of Gisin, Méthot, and Scarani [25] and Joshi et al. [26]. For every $m \geq 200$, the explicit comparison (6.15) shows that uniform-input T_m has greater strength than Q_{2^m} after optimization of the latter's input law.

The projective proof reduces the measurements to fourth roots of the identity while preserving the original strategy as a reducing summand. A matrix Hardy inequality then supplies positive forms indexed by partitions of the circle. A tilted partition law matches the CGLMP coefficients; polynomial bounds control the remaining error. Sections 3–4 develop this argument, Section 5 proves rigidity and stability, and Section 6 treats the operational comparisons. The appendices contain the facet analysis and the certificate data.

2. Preliminaries and notation

There are two parties, inputs $x, y \in \{0,1\}$, and outcomes $a, b \in \mathbb{Z}_d$, with $d \geq 2$. The local polytope $\mathcal{C}_d$ is the convex hull of

$$q_s(a,b|x,y) = \mathbf{1}_{\{a=a_x,\ b=b_y\}}, \qquad s = (a_0, a_1, b_0, b_1) \in \mathbb{Z}_d^4.$$

The no-signalling polytope $\mathcal{P}_d$ consists of nonnegative normalized tables whose one-party marginals are independent of the other party's input.

Write $[u]_d \in \{0, \ldots, d-1\}$ for reduction modulo d, and orient the four contexts cyclically:

$$r_{00} = [a-b]_d, \quad r_{10} = [b-a]_d, \quad r_{11} = [a-b]_d, \quad r_{01} = [b-a-1]_d.$$

Our normalization is

$$S_d(q) = \sum_{x,y} \mathbb{E}_q r_{xy}, \qquad I_d(q) = 4 - \frac{2S_d(q)}{d-1}. \tag{2.1}$$

For a deterministic strategy, the residuals sum to -1 modulo d, so $S_d \geq d-1$, equivalently $I_d \leq 2$ [3, 15].

For the maximally entangled state,

$$|\Phi_n\rangle = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} |jj\rangle, \qquad \tau_n = \mathrm{Tr}\,/n, \qquad \|X\|_{2,n}^2 = \tau_n(X^\dagger X).$$

With $P^0 = A_0^T, P^1 = B_0, P^2 = A_1^T, P^3 = B_1$,

$$q(a,b|x,y) = \tau_n(P_a^{2x} P_b^{1+2y}).$$

PVMs may contain zero projections and unequal ranks. Common conjugation of the four tracial PVMs by U corresponds physically to $\overline{U} \otimes U$, which preserves Φ_n. Operator norms are denoted by $\|\cdot\|$, and $\|\cdot\|_F$ is the unnormalized Frobenius norm. Matrix Hilbert–Schmidt norms and inner products without a dimension subscript use the normalized trace of the matrix space in question.

The standard tracial PVMs on $\mathbb{C}^d$ are

$$|\psi_{t,a}\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} e^{-2\pi \mathrm{i} j(a-t/4)/d} |j\rangle, \qquad \Pi_a^t = |\psi_{t,a}\rangle\langle\psi_{t,a}|, \quad 0 \leq t < 4. \tag{2.2}$$

Alice's physical bases are conjugated and Bob's are unchanged; individual basis-vector phases leave the projectors unchanged. For $n = dk$, define

$$\mathrm{Err}_{d,n}(P) = \inf_{U \in U(n)} \sum_{t=0}^{3} \sum_{a=0}^{d-1} \left\| P_a^t - U(\Pi_a^t \otimes I_k) U^\dagger \right\|_{2,n}^2. \tag{2.3}$$

The stability constants are

$$\begin{aligned} C_d^{\mathrm{st}} &= 2[1+4(1+2\sqrt{d})(1+2\sqrt{d-1})]^2, \\ R_d^{\mathrm{st}} &= C_d^{\mathrm{st}}\left(\sqrt{224}+\frac{8d}{\sqrt{d-1}}\right)^2, \\ K_d &= \frac{3}{4}d(d-1)(d+1)R_d^{\mathrm{st}}. \end{aligned} \tag{2.4}$$

For the enlarged-space constructions, use $\mathsf{C}_a^{(r)} = |a\rangle\langle a| \otimes I_r$ on $\mathbb{C}^d \otimes \mathbb{C}^r$. The symbols $E_{j\ell}$ denote spectral projections of fourth-root unitaries $V_j = \sum_{\ell=0}^{3} \mathrm{i}^\ell E_{j\ell}$. We reserve $\mathcal{Z}$ for the reducing isometry and $\Lambda = \mathcal{Z}\mathcal{Z}^\dagger$ for its range projection. The isometry adjoining the spectral-outcome labels is $\mathcal{V}$, with coordinate projections $\mathsf{D}_{j\ell}$ on the label ancilla. The errors used in Section 5 are

$$\varepsilon_{\mathrm{anc}} = \frac{1}{r}\sum_{j,\ell} \|\mathsf{D}_{j\ell}\mathcal{V} - \mathcal{V}E_{j\ell}\|_F^2\,, \qquad \varepsilon_{\mathrm{red}} = \left\|\Lambda - I_d \otimes \bar{\Lambda}\right\|_{2,N}^2,$$

where $\Lambda = \mathrm{diag}(\Lambda_0, \ldots, \Lambda_{d-1})$, $\bar{\Lambda} = d^{-1}\sum_a \Lambda_a$, and $N = 4dn$. For PVMs P, Q on $\mathbb{C}^n$, write $\varepsilon_{\mathrm{meas}}(P,Q) = \sum_{t,a} \left\|P_a^t - Q_a^t\right\|_{2,n}^2$.

Let $u_d(a,b|x,y) = d^{-2}$. Global critical visibility is

$$v_{\mathrm{crit}}(q) = \max\{v \in [0,1] : vq + (1-v)u_d \in \mathcal{C}_d\}.$$

Smaller visibility means greater white-noise resistance. For the CGLMP functional alone, the corresponding threshold is the CGLMP critical visibility $v_{\mathrm{CGLMP}}(q) = 2/I_d(q)$, the smallest visibility at which the isotropic mixture still violates the CGLMP inequality. For equal-rank PVMs on Φ_n, this is the visibility of the isotropic state $v|\Phi_n\rangle\langle\Phi_n| + (1-v)I/n^2$.

An input distribution μ may correlate the two parties' inputs, and is independent of the hidden variable. Statistical strength in bits per complete trial is

$$\mathcal{S}_\mu(q) = \min_{\ell \in \mathcal{C}_d} \sum_{x,y} \mu_{xy} D(q_{xy}\|\ell_{xy}), \qquad D(p\|q) = \sum_z p_z \log_2(p_z/q_z). \tag{2.5}$$

We use the usual extended-value conventions at zero probabilities. The local fraction and the input-law function are

$$\mathrm{LF}(q) = \max\{\alpha : q \geq \alpha L \text{ entrywise for a normalized } L \in \mathcal{C}_d\},$$

$$\Xi(\mu) = \min_{\substack{0 \leq s_e \leq 1 \\ \sum_e s_e = 3}} -\sum_e \mu_e \log_2 s_e, \qquad e \in \{00, 01, 10, 11\}. \tag{2.6}$$

For $\alpha < 1$, the normalized remainder of $q - \alpha L$ is no-signalling. We also use

$$\mathrm{TV}(p,q) = \frac{1}{2}\sum_z |p_z - q_z|, \qquad H(p,q)^2 = \sum_z (\sqrt{p_z} - \sqrt{q_z})^2,$$

so that $\mathrm{TV}(p,q) \leq H(p,q)$.

3. Fourth roots and the matrix Hardy inequality

3.1. **The companion representation.** For $V_0, \ldots, V_{d-1} \in U(r)$ with $V_j^4 = I$, put

$$z_{jk} = \tau_r(V_j^\dagger V_k), \qquad p_{jk} = \frac{1 - \Re z_{jk} + \Im z_{jk}}{2}, \qquad \mathfrak{F}_d = \sum_{j<k} p_{jk}.$$

For a coefficient sequence $c = (c_h)_{h=1}^{d-1}$, define

$$L_c = \sum_{j<k} \big[c_{k-j}(1 - \Re z_{jk}) + c_{d-k+j} \Im z_{jk} \big]. \tag{3.1}$$

The spectral projections satisfy

$$p_{jk} = \sum_{\ell=0}^{3} \sum_{u=1}^{2} \tau_r(E_{j\ell} E_{k,\ell+u}) \geq 0, \tag{3.2}$$

with spectral indices read modulo four. For $f(x) = \sec(\pi x/2)$, set

$$P_{d,h}(x) = \binom{d}{h} x^h (1-x)^{d-h}, \qquad B_d g(h) = d \int_0^1 P_{d,h}(x) g(x)\, \mathrm{d}x,$$

$$\beta_h = \frac{f(h/d)}{2d^2}, \qquad q_h = \frac{B_d f(h)}{2d^2}, \qquad \mathcal{G}_d = L_\beta, \qquad \mathcal{D}_d = L_q.$$

Lemma 3.1 (Companion representation)**.** *Every projective strategy in Theorem 1.1 has a companion fourth-root tuple of size $r = 4n$ satisfying*

$$I_d^{\mathrm{F}} - I_d = \frac{8d}{d-1} \mathcal{G}_d. \tag{3.3}$$

The associated physical PVMs act on $\mathbb{C}^d \otimes \mathbb{C}^{4n}$, and the original four PVMs form a common reducing summand.

Proof. Let $J|a\rangle = |a-1\rangle$ on $\mathbb{C}^d$, and define

$$Z_t = \sum_{c=0}^{d-1} J^c \otimes P_c^t \quad (0 \leq t < 4), \qquad Z_4 = Z_0 (J \otimes I_n).$$

Each Z_t is unitary and commutes with $J \otimes I_n$, and

$$Z_t \mathsf{C}_a^{(n)} Z_t^\dagger = \sum_{k=0}^{d-1} |k\rangle\langle k| \otimes P_{a-k}^t.$$

This outcome-covariant strategy preserves the Bell value and contains the original strategy in its $k = 0$ block.

Put $\mathsf{U}_t = Z_t^\dagger Z_{t+1}$. Their product is $J \otimes I_n$. On a four-dimensional clock, define

$$\mathcal{R} = \sum_{t=0}^{3} |t\rangle\langle t+1 \bmod 4| \otimes \mathsf{U}_t.$$

Every cyclic product of the U_t's equals $J \otimes I_n$; hence $\mathcal{R}^4 = I_4 \otimes J \otimes I_n$. With

$$\mathsf{F}_{aj} = d^{-1/2} e^{2\pi \mathrm{i} a j/d}, \qquad \lambda_j = e^{\pi \mathrm{i} j/(2d)},$$

reordering tensor factors and Fourier transforming the outcome coordinate gives

$$\mathcal{R} = (\mathsf{F} \otimes I_{4n}) \operatorname{diag}(\lambda_j V_j)_{j=0}^{d-1} (\mathsf{F}^\dagger \otimes I_{4n}), \qquad V_j \in U(4n), \quad V_j^4 = I.$$

In these coordinates the PVMs are

$$\mathcal{M}_a^t(V) = \mathcal{R}^t \mathsf{C}_a^{(4n)} \mathcal{R}^{-t}, \qquad 0 \leq t < 4. \tag{3.4}$$

Clock averaging identifies their four-edge cost with the sum of the original transition costs. In clock zero, $\mathsf{U}_0 \cdots \mathsf{U}_{t-1} = Z_0^\dagger Z_t$. The isometry

$$\mathcal{Z}\xi = \sum_a |a\rangle \otimes |0\rangle \otimes P_a^0 \xi \tag{3.5}$$

therefore satisfies

$$\mathcal{Z}^\dagger \mathcal{Z} = I_n, \qquad \mathcal{M}_a^t(V)\mathcal{Z} = \mathcal{Z}P_a^t.$$

Self-adjointness makes its image a reducing subspace.

The one-edge cost is

$$\frac{d-1}{2} - \frac{1}{2d}\sum_{j<k}\left[\sec\left(\frac{\pi(k-j)}{2d}\right)\Re z_{jk} - \csc\left(\frac{\pi(k-j)}{2d}\right)\Im z_{jk}\right],$$

using $\sum_{a=0}^{d-1} a\omega^{ah} = -d/(1-\omega^h)$ for $h \not\equiv 0 \pmod d$. Subtracting its value at $V_j = I$ gives $d\mathcal{G}_d$. The four edges and (2.1) yield (3.3). For $V_j = I$, Fourier conjugation gives (2.2), and the same sum evaluates its value as (1.1). □

3.2. **Trace-density moments and pinching.** Let P_- retain the strictly negative Fourier coefficients on $\mathbb{T}$. The L^2-norm uses normalized Haar measure and normalized matrix trace.

Theorem 3.2 (Fourth-root Hardy bound)**.** *For every measurable $U : \mathbb{T} \to U(r)$ satisfying $U(z)^4 = z^{-1}I$ almost everywhere,*

$$\|P_- U\|_2^2 \geq \nu_4 := \frac{1}{2} - \frac{4G_{\mathrm{Cat}}}{\pi^2}, \qquad G_{\mathrm{Cat}} = \sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)^2}, \tag{3.6}$$

where G_{Cat} is Catalan's constant.

The moment identity below was obtained independently by Heinävaara [27, Lemma 2.5, equation (16)]. We give a proof adapted to matrix pencils.

Lemma 3.3 (Trace-density moments)**.** *Let $H = H^\dagger$ be invertible and $Y = Y^\dagger$ in $M_r(\mathbb{C})$. Counting algebraic multiplicities, define*

$$w_{H,Y}(y) = \frac{1}{2\pi r}\sum_{\lambda \in \mathrm{spec}(H^{-1}(Y - yI))} |\Im\lambda|.$$

This function is continuous, nonnegative, and supported on the spectral interval of Y. For every integer $m \geq 0$,

$$\int y^m w_{H,Y}(y)\,\mathrm{d}y = \frac{1}{(m+1)(m+2)!}\,\frac{\mathrm{d}^{m+2}}{\mathrm{d}t^{m+2}}\tau_r(H+tY)_+^{m+1}\Big|_{t=0}. \tag{3.7}$$

Proof. For $\Im z > 0$, an eigenvector of $L(z) = H^{-1}(Y - zI)$ satisfies

$$-\Im z\ \|v\|^2 = \Im\lambda\, v^\dagger H v.$$

Its spectrum therefore avoids the real axis. The sum

$$g(z) = r^{-1}\sum_{\Im\lambda(L(z))>0}\lambda(L(z))$$

is holomorphic, being the trace of $L(z)$ times its upper-cluster Riesz projection, with algebraic multiplicities also at repeated or defective eigenvalues.

The sum extends continuously to the real axis. Choose disjoint small disks around the limiting eigenvalues. In a connected upper half-neighbourhood, the number of selected roots in each disk is constant. A real limiting cluster contributes an integer multiple of its real eigenvalue. At real y, the characteristic polynomial $\det(Y - yI - \lambda H)/\det H$ has real coefficients, so nonreal roots occur in conjugate pairs. Consequently

$$\Im g(y + \mathrm{i}0) = \pi w_{H,Y}(y).$$

Outside the spectral interval of Y, definiteness of $Y - yI$ makes the pencil similar, up to sign, to a Hermitian matrix, and the density vanishes.

Near $w = 0$, put

$$K(w) = H^{-1}(I - wY), \qquad q(w) = \tau_r(K(w)\Pi_-(w)),$$

where $\Pi_-(w)$ continues the negative spectral cluster of H^{-1}. For small real w, similarity to $(I - wY)^{1/2}H^{-1}(I - wY)^{1/2}$ shows that q has real Taylor coefficients. In the upper exterior domain, $g(z) = -zq(1/z)$. Writing $q(w) = \sum_j q_j w^j$, Schwarz reflection and Cauchy's formula around the spectral slit give

$$g(z) = -q_0 z - q_1 + \int \frac{w_{H,Y}(y)}{y - z}\,\mathrm{d}y, \qquad \int y^m w_{H,Y}(y)\,\mathrm{d}y = [w^{m+2}]q(w). \tag{3.8}$$

The endpoint arcs vanish because pencil eigenvalues remain bounded near finite points; continuity controls the two banks of the slit.

Let $k = m + 2$ and $\mathsf{R}_H(\lambda) = (\lambda I - H)^{-1}$. Contours enclose the indicated positive or negative spectra and exclude zero. Expanding the logarithmic determinant ratio gives

$$\begin{aligned}
[w^k]q(w) &= \frac{(-1)^k}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H^{-1},-}} \operatorname{Tr}[((\zeta H - I)^{-1}Y)^k]\,\mathrm{d}\zeta \\
&= -\frac{1}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H,-}} \lambda^{k-2} \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k]\,\mathrm{d}\lambda \\
&= \frac{1}{kr\,2\pi\mathrm{i}} \oint_{\Gamma_{H,+}} \lambda^{k-2} \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k]\,\mathrm{d}\lambda.
\end{aligned}$$

The second line substitutes $\lambda = 1/\zeta$; the third uses decay $O(\lambda^{-2})$ at infinity. Positive-cluster functional calculus and integration by parts, with

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k] = -k \operatorname{Tr}[(\mathsf{R}_H(\lambda)Y)^k \mathsf{R}_H(\lambda)],$$

give

$$[t^k]\tau_r(H + tY)_+^{k-1} = (k - 1)[w^k]q(w).$$

Together with (3.8), this proves (3.7).

For the two-node specialization, expansion at a gives

$$\operatorname*{Res}_{\lambda=a} \frac{\lambda^{m-2}}{(\lambda - a)^p(\lambda - b)^q} = \binom{m-2}{p-1} \frac{a^{q-1}(-b)^{p-1}}{(a - b)^{m-1}}, \qquad p, q \geq 1, \quad p + q = m, \quad a > 0 > b. \tag{3.9}$$

□

Lemma 3.4 (Positive pinching identity)**.** *Let $B = B^\dagger$, $B^2 = I$, $Y = Y^\dagger$, and*

$$C = \Pi_B(Y) := \frac{Y + BYB}{2}, \qquad D = Y - C.$$

There is a nonnegative finite measure ν, supported on $[-1, 1]$ times the spectral interval of Y, such that

$$\tau_r f(sB + tY) - \tau_r f(sB + tC) = t^2 \int f''(-sx + ty)\,\mathrm{d}\nu(x, y) \tag{3.10}$$

for polynomials f. Its density on $|x| < 1$ is

$$\frac{\omega(x, y)}{1 - x^2}, \qquad \omega(x, y) = \frac{1}{2\pi r} \sum_{\lambda \in \operatorname{spec}((B - xI)(Y - yI))} |\Im\lambda|, \tag{3.11}$$

and

$$0 \leq \frac{\omega(x, y)}{1 - x^2} \leq \frac{\|D\|}{2\pi\sqrt{1 - x^2}}, \qquad \nu(\mathbb{R}^2) = \frac{1}{2}\tau_r(D^2). \tag{3.12}$$

Proof. Apply Lemma 3.3 with $H = (B - xI)^{-1}$. In a basis diagonalizing B, let p entries of an index cycle $\mathbf{i} = (i_1, \dots, i_m)$ belong to its $+1$ eigenspace and let $q = m - p$. Put $Y_{\mathbf{i}} = Y_{i_1 i_2} \cdots Y_{i_m i_1}$. Substituting $a = (1 - x)^{-1}$, $b = -(1 + x)^{-1}$ in (3.9) yields

$$\int y^{m-2} \frac{\omega(x, y)}{1 - x^2} \, \mathrm{d}y = \frac{1}{rm \, 2^{m-1}} \sum_{\mathbf{i}:\, 1 \le p < m} \binom{m-2}{p-1} (1 - x)^{p-1} (1 + x)^{q-1} Y_{\mathbf{i}}. \tag{3.13}$$

The coefficient of t^m in the mixed-cycle part of $\tau_r e^{sB + tY}$ is

$$\frac{1}{rm} \sum_{\mathbf{i}:\, 1 \le p < m} Y_{\mathbf{i}} \int_{\substack{u_j \ge 0 \\ \sum_j u_j = 1}} e^{s \sum_j b_{i_j} u_j} \, \mathrm{d}u.$$

Grouping the positive-B coordinates with sum v gives a beta integral with density $v^{p-1}(1 - v)^{q-1}/((p - 1)!(q - 1)!)$. The substitution $x = 1 - 2v$, followed by (3.13), gives the coefficient on the right of (3.10), including the sign in $-sx$. Comparison of coefficients proves the polynomial identity.

A block-diagonal similarity transforms the pencil in (3.11) into a matrix with Hermitian diagonal blocks and off-diagonal blocks

$$\sqrt{1 - x^2} D_{+-}, \qquad -\sqrt{1 - x^2} D_{+-}^{\dagger}.$$

Its imaginary part has norm at most $\sqrt{1 - x^2}\, \|D\|$. The same bound holds for the imaginary part of each eigenvalue, proving the majorant in (3.12). Integration of the $m = 2$ case of (3.13) in x gives the total mass. □

Corollary 3.5 (Holomorphic pinching on a strip)**.** *Under the hypotheses of Lemma 3.4, identity* (3.10) *holds with $s = \rho \in [0, 1)$ and $t = \mathrm{i}$ for every f holomorphic on $|\Re z| < 1$.*

Proof. Choose $\rho < \rho' < 1$ and $L > \|Y\|$. The rectangle $\{z : |\Re z| \le \rho',\ |\Im z| \le L\}$ contains the spectra of $\rho B + \mathrm{i}Y, \rho B + \mathrm{i}C$ and all scalar arguments $-\rho x + \mathrm{i}y$ in the support of ν. Sum the polynomial identity for a large-parameter resolvent, continue in the connected exterior of this rectangle, and apply Cauchy's formula on its boundary. □

3.3. **Proof of the Hardy bound.**

Proof of Theorem 3.2. On $|\Re z| < 1$, define h by

$$h(0) = \frac{8G_{\mathrm{Cat}}}{\pi^2}, \qquad h'(0) = 0, \qquad h''(z) = -\sec(\pi z/2).$$

Its real part $u(x, y) = \Re h(x + \mathrm{i}y)$ extends continuously to the closed strip and satisfies

$$u(\pm 1, y) = |y|, \qquad 0 \le u(x, y) \le \frac{8G_{\mathrm{Cat}}}{\pi^2} + |y|. \tag{3.14}$$

Indeed,

$$h'(z) = -\frac{2}{\pi} \log\bigl(\sec(\pi z/2) + \tan(\pi z/2)\bigr).$$

The boundary imaginary parts give the slopes of $|y|$, and $\int_0^1 (1 - x) \sec(\pi x/2) \, \mathrm{d}x = 8G_{\mathrm{Cat}}/\pi^2$ fixes the constant. Also $u_{xx} = -\Re \sec(\pi(x + \mathrm{i}y)/2) < 0$. Symmetry in x and integration of $\operatorname{sech}(\pi y/2)$ give (3.14).

For a matrix X with spectrum in the closed strip, put

$$\mathscr{U}(X) = r^{-1} \sum_{\lambda \in \operatorname{spec} X} \Re h(\lambda).$$

Apply Corollary 3.5 and let $\rho \uparrow 1$. Since $\Pi_B(Y)$ commutes with B, the boundary values in (3.14) give

$$\mathscr{U}(B+\mathrm{i}Y) - \tau_r|\Pi_B(Y)| = \int \Re \sec\left(\frac{\pi(x+\mathrm{i}y)}{2}\right) \mathrm{d}\nu(x,y) \geq 0. \tag{3.15}$$

For the limit, combine (3.12) with

$$\int_{\mathbb{R}} \Re \sec\left(\frac{\pi(\rho x+\mathrm{i}y)}{2}\right) \mathrm{d}y = 2.$$

The absolute integral of each x-slice is at most $\|D\|/(\pi\sqrt{1-x^2})$. Restriction to smaller strips followed by passage to the endpoints proves convergence for the possibly nonnormal matrix $B+\mathrm{i}Y$.

Set $\mathcal{W}(s) = sU(s^4)$ on the circle and

$$A = \Re\mathcal{W} + \Im\mathcal{W}, \qquad B = \Re\mathcal{W} - \Im\mathcal{W}.$$

These are commuting Hermitian involutions. The identity $\mathcal{W}(\mathrm{i}s) = \mathrm{i}\mathcal{W}(s)$ makes their means zero. For the circular Hilbert transform with multiplier $-\mathrm{i}\,\mathrm{sgn}(k)$, Parseval gives

$$1 - 2\,\|P_-U\|_2^2 = \int_{\mathbb{T}} \tau_r(A\,\mathsf{H}B)\,\mathrm{d}m \leq \int_{\mathbb{T}} \tau_r|\Pi_B(\mathsf{H}B)|\,\mathrm{d}m. \tag{3.16}$$

The frequencies of $\mathcal{W}$ are $4k+1$, with the same positive/negative split as those of U. Expanding $\mathcal{W} = ((1+\mathrm{i})A + (1-\mathrm{i})B)/2$ proves the equality.

The analytic matrix function with boundary values $B + \mathrm{i}\mathsf{H}B$ has value zero at the disk centre. Trace holomorphic functional calculus and the mean-value identity give mean $\mathscr{U} = 8G_{\mathrm{Cat}}/\pi^2$. For boundary passage, use radial limits and

$$0 \leq \mathscr{U}(X) \leq \frac{8G_{\mathrm{Cat}}}{\pi^2} + \tau_r|\Im X|.$$

This follows from (3.14), Schur triangularization, and convexity of absolute value on the diagonal. The radial Hilbert transforms are uniformly L^2-bounded, hence the displayed quantities are uniformly integrable. Applying (3.15) in (3.16) gives $1 - 2\,\|P_-U\|_2^2 \leq 8G_{\mathrm{Cat}}/\pi^2$, proving (3.6). □

4. Positive partition forms and the CGLMP bound

Theorem 4.1 (Fourth-root bound)**.** *Let $d \geq 2$, $r \geq 1$, and $V_0, \dots, V_{d-1} \in U(r)$ satisfy $V_j^4 = I$. Then*

$$\mathcal{G}_d \geq \left(\frac{3}{2} - \frac{\pi}{4}\right)\mathcal{D}_d + \frac{\mathfrak{F}_d}{6d^2(d+1)}, \qquad \mathcal{D}_d \geq 0. \tag{4.1}$$

4.1. From Hardy space to circular partitions.

Lemma 4.2 (Partition forms)**.** *For a consecutive partition $\ell = (I_0, \dots, I_{d-1})$ of $[0,1]$, the fourth-root tuple defines a nonnegative form*

$$\begin{aligned} \mathcal{K}_\ell(V) &:= \left\| P_-[e^{-\pi\mathrm{i}t/2}V(t)^\dagger] \right\|_2^2 - \nu_4 \\ &= \frac{1}{2}\sum_{j<k} \int_{I_j}\int_{I_k} [f(t-s)(1-\Re z_{jk}) + f(1-t+s)\Im z_{jk}]\,\mathrm{d}t\,\mathrm{d}s \geq 0, \end{aligned} \tag{4.2}$$

where $V(t) = V_j$ on I_j. Extend $V_{j+d} = -\mathrm{i}V_j$. Averaging over the d cyclic assignments to a circular partition gives L_c, with

$$c_h = \frac{1}{2d}\int_0^1 f(x)\,\Pr(N_x = h)\,\mathrm{d}x, \qquad 1 \leq h < d, \tag{4.3}$$

where N_x counts boundaries in a length-x arc from a uniform origin. In particular, $\mathcal{D}_d \geq 0$.

Proof. The function $U(e^{2\pi it}) = e^{-\pi it/2}V(t)^\dagger$ satisfies $U(z)^4 = z^{-1}I$, so (3.6) gives positivity. For $s < t$, put $x = t - s$ and use

$$e^{\pi ix/2}(1 + \mathrm{i}\cot \pi x) = \frac{f(x) + \mathrm{i}f(1-x)}{2}.$$

Abel regularization and pairing the two half-squares in $P_- = (I - P_0 - \mathrm{i}\mathsf{H})/2$ give (4.2), with constant $\frac{1}{2} - \frac{1}{2}\int_0^1 (1-x)f(x)\,\mathrm{d}x = \nu_4$. The integrals are finite: near the diagonal the imaginary part vanishes within an interval, while cross-interval regions have length $O(x)$. This also justifies the regularization limit.

The cyclic average uses the zero Hardy gap of scalar cuts. For the periodic branch $U_0(t) = e^{-\pi it/2}$, $0 \le t < 1$,

$$\|P_- U_0\|_2^2 = \frac{8}{\pi^2}\sum_{n\ge 0}\frac{1}{(4n+3)^2} = \nu_4.$$

A scalar label dropping by one power of i at T gives

$$e^{-\pi it/2}\mathrm{i}^{-a+\mathbf{1}_{\{t>T\}}} = \mathrm{i}^{1-a}e^{-\pi iT/2}U_0([t-T]_1) \quad \text{almost everywhere.} \tag{4.4}$$

Translation and multiplication by a constant phase preserve Fourier moduli, so (4.4) has the same negative Fourier energy.

For a pair of intervals separated by h boundaries, the direct arc contributes $f(x)$ and the complementary arc contributes $f(1-x)$. Crossing the label origin replaces z by $-\mathrm{i}\bar{z}$, interchanging these coefficients. Subtraction of the value at $V_j = I$ removes the scalar-cut contribution, and averaging the starting label gives the factor $1/d$. This proves (4.3). Independent uniform boundaries have $\Pr(N_x = h) = P_{d,h}(x)$, giving $L_c = \mathcal{D}_d \ge 0$. □

4.2. **The tilted law and coefficient matching.** Define

$$v(x) = -\frac{x(1-x)}{2} + \frac{2x-1}{\pi[\sec(\pi x/2) - \csc(\pi x/2)]}, \tag{4.5}$$

with removable singularities filled continuously. Also put

$$b(x) = -\frac{1}{2}\left[x(1-x)f(x) - \frac{2x}{\pi}\right]'', \qquad S(x) = \frac{1 - z\cot z}{\sin^2 z}, \quad z = \frac{\pi}{2}(x - \tfrac{1}{2}), \quad S(1/2) = \frac{1}{3},$$

and define

$$e_{d,h} = f(h/d) - B_d f(h) - \frac{1}{d}B_d b(h), \qquad s_{d,h} = \frac{1}{d}B_d S(h). \tag{4.6}$$

Lemma 4.3 (Tilted partition law)**.** *There is a probability density on circular partitions whose averaged Hardy form satisfies*

$$\mathcal{K}_d^v = L_{\{B_d f + (1/d)B_d(vf)''\}/(2d^2)} \ge c_* \mathcal{D}_d, \qquad c_* = \frac{3}{2} - \frac{\pi}{4}. \tag{4.7}$$

Moreover,

$$\mathcal{G}_d = \mathcal{K}_d^v + \frac{1}{2d^2}L_{e_d + s_d}, \qquad s_{d,h} = s_{d,d-h} \ge \frac{1}{3(d+1)}. \tag{4.8}$$

Proof. The function v is symmetric about $1/2$, has zero value and first derivative at both endpoints, and $v''(0) = v''(1) = (\pi - 2)/2$. It has the expansion

$$v(x) = \sum_{n\ge 1} a_n \frac{\sin^2(\pi n x)}{\pi^2 n^2}, \qquad a_n > 0, \qquad \sum_{n\ge 1} a_n = \frac{\pi}{4} - \frac{1}{2}. \tag{4.9}$$

To prove this, write $u = x - \frac{1}{2}$, so that

$$v(x) = \frac{u^2 - \frac{1}{4}}{2} + \frac{u}{\sqrt{2}\pi}\left[\csc(\pi u/2) - 2\sin(\pi u/2)\right].$$

The nonconstant even coefficients of $u\csc(\pi u/2)$ are positive. For $a = \pi/2$,

$$-[u\sin(au)]^{(4)} = a^3[4\cos(au) - au\sin(au)] > 0 \quad (|au| \le \pi/4).$$

Thus $v^{(4)} > 0$, and four integrations by parts give

$$a_n = \frac{1}{4\pi^2 n^2}\int_0^1 v^{(4)}(x)[1 - \cos(2\pi n x)]\,\mathrm{d}x > 0.$$

The coefficients are $O(n^{-2})$. Fourier inversion and evaluation of $v''(0)$ prove (4.9).

Tilt independent uniform boundary points $T_1, \dots, T_d$ by

$$\varpi_d = 1 + \frac{1}{d}\sum_{a<b} v''(T_a - T_b) = c_* + \frac{1}{d}\sum_{n\ge 1} a_n \left|\sum_{j=1}^{d} e^{2\pi i n T_j}\right|^2, \tag{4.10}$$

using the periodic extension in the differences. Its expectation is one and $\varpi_d \ge c_* > 0$. For one Fourier-power density, the count generating function is

$$(1 - x + xz)^d + (d-1)\frac{\sin^2(\pi n x)}{\pi^2 n^2}(z-1)^2(1 - x + xz)^{d-2}.$$

Summing with (4.9) shows that under (4.10),

$$\Pr_{\varpi_d}(N_x = h) = P_{d,h}(x) + \frac{v(x)}{d}P''_{d,h}(x). \tag{4.11}$$

Equations (4.3) and (4.11), followed by two integrations by parts, give (4.7). The endpoint terms vanish because vf vanishes at both endpoints, $(vf)'$ has finite one-sided limits, and $P_{d,h}$ vanishes there.

Differentiating (4.5) gives $(vf)'' = b - S$. The Taylor series of $z\cot z$ gives $S(x) = S(1-x) \ge 1/3$. Substitution in (4.6) proves (4.8). □

4.3. **Circulation and the scalar estimate.**

Lemma 4.4 (Circulation bound)**.** *For $i < j < k$, let $C_{ijk} = \Im z_{ij} + \Im z_{jk} - \Im z_{ik}$. Then*

$$|C_{ijk}| \le 5(p_{ij} + p_{jk} + p_{ik}). \tag{4.12}$$

For $t_h = e_{d,d-h} - e_{d,h}$ and $\omega_{ijk} = t_{j-i} + t_{k-j} - t_{k-i}$,

$$L_e = 2\sum_{j<k} e_{d,k-j}p_{jk} + \frac{1}{d}\sum_{i<j<k}\omega_{ijk}C_{ijk}. \tag{4.13}$$

Proof. The sum-of-squares identity (B.1) in Appendix B proves (4.12). The skew edge array t has zero divergence. Expanding (3.1) over triangles then gives (4.13). □

The absolute triangle load at separation h is

$$\ell_{d,h} = \sum_{k=1}^{h-1}|t_k + t_{h-k} - t_h| + \sum_{k=h+1}^{d-1}|t_h + t_{k-h} - t_k|. \tag{4.14}$$

For $b_d = \max_h |e_{d,h}|$, the bounds $|\omega_{ijk}| \le 6b_d$ and (4.12) give

$$L_e \ge -32 b_d \mathfrak{F}_d, \tag{4.15}$$

since each edge belongs to $d-2$ triangles.

Proof of Theorem 4.1. Lemma 4.2 gives $\mathcal{D}_d \geq 0$. Use the normalized scalar functions $\mathcal{E}_\theta$ in Appendix D, whose normalization (D.1) and bound (D.2) hold on the indicated domains. For $d \geq 3$, take

$$\theta = 1/d, \qquad (x_1, x_2, x_3) = \left(\frac{j-i}{d}, \frac{k-j}{d}, \frac{d-k+i}{d} \right).$$

Then

$$\Omega_\theta = -\frac{(d+1)(d-2)}{d}\, \omega_{ijk}.$$

Put $b_d^{\triangle} = \max_{i<j<k} |\omega_{ijk}|$. The independent choice of y in (D.2) gives, for every h,

$$\frac{1}{3(d+1)} + e_{d,h} - \frac{5(d-2)}{2d} b_d^{\triangle} \geq \frac{1}{6(d+1)}.$$

Applying Lemma 4.4 and counting $d-2$ triangles per edge yields

$$L_{e+s} \geq \frac{\mathfrak{F}_d}{3(d+1)}.$$

For $d = 2$, the triangle sum is empty and (D.6) gives the same conclusion. Lemma 4.3, through (4.7) and (4.8), now proves (4.1). □

Remark 4.5 (Alternative coefficient). A beta–Stein expansion for $d \geq 66$, together with an interval calculation for $2 \leq d \leq 65$, gives

$$\mathcal{G}_d \geq c_* \mathcal{D}_d + \eta_d \mathfrak{F}_d, \qquad \eta_d = \begin{cases} [6d^2(d+1)]^{-1}, & 2 \leq d \leq 65, \\ (2d-131)[6d^2(d+1)(d+2)]^{-1}, & d \geq 66. \end{cases} \tag{4.16}$$

In particular, $\eta_d \geq [6d^2(d+1)(d+2)]^{-1}$. The scalar estimates and derivation are given in Appendix C.

5. Rigidity and same-dimension stability

5.1. **Equality and multiplicity.**

Theorem 1.1 (restated). *Let $d \geq 2$, $n \geq 1$, and let the four measurements on Φ_n be d-outcome PVMs. Then $I_d \leq I_d^{\mathrm{F}}$, with I_d^{F} given by* (1.1)*. Equality holds exactly when $n = dk$ and the tracial PVMs are commonly unitarily conjugate to $\Pi_a^t \otimes I_k$. All projective maximizers on Φ_d are rank-one Fourier-phase measurements.*

Proof of Theorem 1.1. Lemma 3.1 and Theorem 4.1 give the bound. Equality forces $\mathfrak{F}_d = 0$, so all forbidden products in (3.2) vanish.

For two spectral PVMs E_a, F_b, the allowed columns of row a are $a, a-1$ modulo four. Distinct rows share at most one allowed column. For $a \neq c$,

$$0 = E_a E_c = \sum_b E_a F_b E_c$$

has at most one potentially nonzero term, which must vanish. Thus the two PVMs commute. Applying this argument to every pair gives simultaneous commutativity.

On each joint scalar eigenspace, the labels change at most once, by decreasing the phase by $\pi/2$. Hence they are cuts:

$$V_j = \mathrm{i}^{a - \mathbf{1}_{\{j > j_0\}}}, \qquad a \in \mathbb{Z}_4, \qquad 0 \leq j_0 < d. \tag{5.1}$$

For $s = d - 1 - j_0$,

$$\lambda_j \mathrm{i}^{-\mathbf{1}_{\{j>j_0\}}} = \lambda_s^{-1} \lambda_{j+s \bmod d}.$$

This frequency shift is implemented by a diagonal unitary in the outcome coordinate. Every cut therefore gives the standard PVMs and zero gap, and a direct sum of cuts gives their standard amplification.

The algebra generated by the standard PVMs is $M_d(\mathbb{C})$. In the gauge where the first PVM is computational, the next basis is $U_{\mathrm{F}} = \mathsf{F}\,\mathrm{diag}(\lambda_j)\mathsf{F}^\dagger$, with

$$(U_{\mathrm{F}})_{ab} = \frac{1-\mathrm{i}}{d[1-e^{2\pi\mathrm{i}(a-b+1/4)/d}]}.$$

Every entry is nonzero. An operator commuting with the first PVM is diagonal; commuting with the rank-one projectors of this second basis forces its diagonal entries to agree. The commutant of an amplification is therefore $I_d \otimes M_k(\mathbb{C})$. Its reducing projections have ranks divisible by d, and every restriction is a standard amplification. Apply this to the reducing isometry (3.5). It gives $d \mid n$ and the classification in the original dimension. The converse follows from (2.2). □

5.2. **Rounding spectral measurements.** The constants in (2.4) arise in two stages. Rounding and descent give C_d^{st} and R_d^{st} in terms of $\mathfrak{F}_d$. Conversion to a Bell-value deficit uses the coefficient $1/[6d^2(d+1)]$ in (4.1), and supplies K_d, the arithmetic bound, the rank-forcing threshold, and $\widehat{K}_d$ in (6.17).

Lemma 5.1 (Rounding to cuts)**.** *For a fourth-root tuple as in Theorem 4.1, the squared distance, summed over sites and spectral outcomes, from $(E_{j\ell})$ to a commuting cut tuple on the same $\mathbb{C}^r$ is at most $C_d^{\mathrm{st}}\mathfrak{F}_d$. For the corresponding fourth-root unitaries D_j,*

$$\sum_j \|V_j - D_j\|_{2,r}^2 \le 2C_d^{\mathrm{st}}\mathfrak{F}_d.$$

Proof. For two q-outcome PVMs whose distinct allowed rows have at most one common allowed column, total forbidden overlap f_{bad} bounds

$$\sum_{\ell,m} \|[E_\ell, F_m]\|_2^2 \le 4q f_{\mathrm{bad}}.$$

For $\ell \neq \ell'$, put $x_m = E_\ell F_m E_{\ell'}$. The identity $\sum_m x_m = 0$ has at most one term with both endpoints allowed. Bound that term by Cauchy–Schwarz in the sum of the others, and sum over ℓ, ℓ', m, using

$$\sum_{\ell,m} \|[E_\ell, F_m]\|_2^2 = 2 \sum_{\ell\neq\ell',m} \|E_\ell F_m E_{\ell'}\|_2^2 .$$

For the present graph this gives

$$\Delta_{\mathrm{com}} := \sum_{j<k,\ell,m} \|[E_{j\ell}, E_{km}]\|_2^2 \le 16\mathfrak{F}_d. \tag{5.2}$$

For m PVMs, coherently adjoin the successive outcome labels:

$$\mathcal{V}\xi = \sum_{\boldsymbol{\ell}} E_{m-1,\ell_{m-1}} \cdots E_{0,\ell_0}\xi \otimes |\boldsymbol{\ell}\rangle.$$

This is an isometry. For the coordinate projections $\mathsf{D}_{j\ell}$, the error defined in the preliminaries obeys

$$\varepsilon_{\mathrm{anc}} \le 2\Delta_{\mathrm{com}}. \tag{5.3}$$

To prove this estimate, let T_i be orthogonal projections, $x_i = T_i x_{i-1}$, and set

$$\epsilon = \sum_i \|(I-T_i)x\|^2, \qquad L = \|x\|^2 - \|x_m\|^2, \qquad D = \|x\|^2 - \Re\langle x, x_m\rangle.$$

Telescoping and Cauchy–Schwarz give $D \le \sqrt{\epsilon L}$, while $L \le 2D$, hence $D \le 2\epsilon$. Apply this on Hilbert–Schmidt space to the pinching maps. The identity

$$\left\| X - \sum_\ell E_\ell X E_\ell \right\|_2^2 = \frac{1}{2} \sum_\ell \|[E_\ell, X]\|_2^2$$

gives (5.3).

Pinch $\mathcal{V}\mathcal{V}^\dagger$ by the joint ancilla projections, obtaining a positive contraction Γ of trace r, with $r^{-1} \operatorname{Tr}(\Gamma - \Gamma^2) \le \varepsilon_{\mathrm{anc}}$. Within the ancilla blocks, choose the projection Γ' onto the r largest eigenvalues. If

$$\zeta_{\mathrm{tail}} = \sum_{i>r} \lambda_i = \sum_{i \le r} (1 - \lambda_i),$$

then

$$\zeta_{\mathrm{tail}} \le \sum_i \lambda_i (1 - \lambda_i), \qquad \left\| \mathcal{V}\mathcal{V}^\dagger - \Gamma' \right\|_F^2 = 2\zeta_{\mathrm{tail}}.$$

A closest isometry $\mathcal{T}$ onto $\operatorname{ran} \Gamma'$ satisfies $r^{-1} \|\mathcal{T} - \mathcal{V}\|_F^2 \le 2\varepsilon_{\mathrm{anc}}$. Compression of the ancilla projections gives commuting PVMs $Q_{j\ell}$ on the original space, satisfying

$$\sum_{j,\ell} \|E_{j\ell} - Q_{j\ell}\|_2^2 \le 2(1 + 2\sqrt{m})^2 \Delta_{\mathrm{com}}. \tag{5.4}$$

Indeed, the square root of the left side is at most $\sqrt{\varepsilon_{\mathrm{anc}}} + \sqrt{2m/r}\, \|\mathcal{T} - \mathcal{V}\|_F$. Zero singular values of the polar overlap are handled by completing the partial isometry between the remaining equal-dimensional spaces.

Take $m = d$, and call the left side of (5.4) g. Completeness and the triangle inequality for concatenated forbidden products give

$$\sqrt{\mathfrak{F}(Q)} \le \sqrt{\mathfrak{F}(E)} + \sqrt{2(d-1)g}.$$

On a joint eigenspace of the commuting PVMs, let b count forbidden pairs in the label string. Delete their left endpoints, at most b positions. The retained labels extend to a cut. Changing one label costs squared PVM distance two, so the repair cost is at most $2\mathfrak{F}(Q)$. Combining with (5.2) gives $C_d^{\mathrm{st}} \mathfrak{F}_d$. Replacing labels by fourth roots multiplies the squared bound by at most two. □

5.3. **Descent and the arithmetic gap.**

Theorem 1.2 (restated). *For d-outcome projective measurements on Φ_n, put $\Delta = I_d^{\mathrm{F}} - I_d$ and $s = n \bmod d$. If $d \mid n$, then*

$$\mathrm{Err}_{d,n}(P) \le \min\{8, K_d \Delta\}, \qquad K_d \le 786432 d^6.$$

For arbitrary n,

$$\Delta \ge \frac{s(d-s)}{3n\, d^4 (d+1) C_d^{\mathrm{st}}},$$

and, for $s > 0$, the projective maximum satisfies

$$\frac{s(d-s)}{3n\, d^4 (d+1) C_d^{\mathrm{st}}} \le I_d^{\mathrm{F}} - \mathcal{I}_{d,n} \le \frac{s}{n} (I_d^{\mathrm{F}} - 2).$$

Proof of Theorem 1.2. Equations (4.1) and (3.3) give, for the companion,

$$\mathfrak{F}_{\mathrm{comp}} \le \frac{3}{4} d(d-1)(d+1)\Delta. \tag{5.5}$$

Fourier unitarity and Lemma 5.1 imply

$$\|\mathcal{R}(V) - \mathcal{R}(D)\|_{2,dr}^2 \le \frac{2C_d^{\mathrm{st}}}{d} \mathfrak{F}_d.$$

For unitaries U_1, U_2,

$$\sum_a \left\| U_1 \mathsf{C}_a^{(r)} U_1^\dagger - U_2 \mathsf{C}_a^{(r)} U_2^\dagger \right\|_2^2 \le 2 \left\| U_1 - U_2 \right\|_2^2 .$$

Telescoping powers in (3.4) gives

$$\begin{aligned} \sum_a \left\| \mathcal{M}_a^t(V) - \mathcal{M}_a^t(D) \right\|_{2,dr}^2 &\le \frac{4t^2 C_d^{\mathrm{st}}}{d} \mathfrak{F}_d, \\ \sum_{t,a} \left\| \mathcal{M}_a^t(V) - \mathcal{M}_a^t(D) \right\|_{2,dr}^2 &\le \frac{56 C_d^{\mathrm{st}}}{d} \mathfrak{F}_d. \end{aligned} \tag{5.6}$$

Standardize the cut benchmark by the controlled diagonal unitary following (5.1). It commutes with the computational PVM. Expressing $\mathcal{Z}$ in this gauge, its range projection is

$$\Lambda = \mathcal{Z}\mathcal{Z}^\dagger = \mathrm{diag}(\Lambda_0, \dots, \Lambda_{d-1}), \qquad \operatorname{rank} \Lambda = n, \qquad N = 4dn.$$

Use $\bar{\Lambda} = d^{-1} \sum_a \Lambda_a$ and $\varepsilon_{\mathrm{red}}$ as defined in the preliminaries. For the second standard basis, let $B_{ab} = |(U_{\mathrm{F}})_{ab}|^2$. Its Fourier eigenvalues are

$$\widehat{B}_k = e^{\pi \mathrm{i} k/(2d)} \left(1 - \frac{k}{d} - \mathrm{i} \frac{k}{d} \right),$$

so BB^T has eigenvalues $1 - 2k(d-k)/d^2$. For its standard projectors Q_b, block expansion gives

$$\sum_b \left\| [\Lambda, Q_b] \right\|_{2,N}^2 \ge \frac{4(d-1)}{d^2} \varepsilon_{\mathrm{red}}.$$

The projection Λ commutes with the actual measurements. Using (5.6) at $t = 1$ gives

$$\varepsilon_{\mathrm{red}} \le \frac{d C_d^{\mathrm{st}}}{d-1} \mathfrak{F}_{\mathrm{comp}}. \tag{5.7}$$

Moreover,

$$0 \le \bar{\Lambda} \le I, \qquad \operatorname{Tr} \bar{\Lambda} = n/d, \qquad \varepsilon_{\mathrm{red}} = \frac{1}{4n} \operatorname{Tr}(\bar{\Lambda} - \bar{\Lambda}^2).$$

A positive contraction of trace $k + f$, $0 \le f < 1$, has $\operatorname{Tr}(\bar{\Lambda} - \bar{\Lambda}^2) \ge f(1-f)$: moving eigenvalue mass to endpoints maximizes $\operatorname{Tr} \bar{\Lambda}^2$ and leaves at most one fractional eigenvalue. Taking $f = s/d$ and applying (5.7) and (5.5) proves (1.3).

If $n = dk$, choose the rank-k projection $\bar{\Lambda}'$ onto the largest eigenvalues of $\bar{\Lambda}$, and put $\Lambda' = I_d \otimes \bar{\Lambda}'$. The rank-repair estimate gives $\|\Lambda - \Lambda'\|_{2,N}^2 \le 2\varepsilon_{\mathrm{red}}$. A closest isometry $\mathcal{J} : \mathbb{C}^n \to \operatorname{ran} \Lambda'$ satisfies

$$n^{-1} \left\| \mathcal{J} - \mathcal{Z} \right\|_F^2 \le 8d \varepsilon_{\mathrm{red}}.$$

The standard projectors preserve this range. Writing $\mathcal{M}_a^t = \mathcal{M}_a^t(V)$, their intertwining error is

$$Q_a^t \mathcal{J} - \mathcal{J} P_a^t = (Q_a^t - \mathcal{M}_a^t)\mathcal{Z} + Q_a^t(\mathcal{J} - \mathcal{Z}) - (\mathcal{J} - \mathcal{Z}) P_a^t.$$

Its total square-root norm is at most

$$\sqrt{224 C_d^{\mathrm{st}} \mathfrak{F}_{\mathrm{comp}}} + 8\sqrt{d \varepsilon_{\mathrm{red}}}.$$

Equations (5.7) and (5.5) prove (1.2). The direct bound is 8, two per PVM. Finally,

$$C_d^{\mathrm{st}} \le 2048 d^2, \qquad \left(\sqrt{224} + \frac{8d}{\sqrt{d-1}} \right)^2 < 512 d,$$

giving $K_d \le 786432 d^6$.

For the upper bound in (1.4), use $\lfloor n/d \rfloor$ standard blocks and s one-dimensional deterministic blocks of value 2. The maximum exists by compactness of the finite-dimensional PVM space. □

For each fixed nonzero residue class modulo d, (1.4) gives $I_d^{\mathrm{F}} - \mathcal{I}_{d,n} = \Theta(1/n)$.

Corollary 5.2 (Rank forcing)**.** *For $d \mid n$, a projective strategy on Φ_n satisfying $\Delta < 2/(nK_d)$ has every outcome rank equal to n/d.*

Proof. For projections, $\mathrm{Tr}(P-Q)^2 \geq |\operatorname{rank} P - \operatorname{rank} Q|$. Any nonconstant rank pattern has total discrepancy at least two. Theorem 1.2 gives the result. □

6. Noise robustness and statistical strength

6.1. **A local component of the standard behaviour.** Geometric summation of (2.2) gives

$$Q_d(a,b|x,y) = \frac{f_d(r_{xy})}{d}, \qquad f_d(r) = \frac{1}{2d^2 \sin^2(\pi(r+1/4)/d)}. \tag{6.1}$$

Construct L_d by choosing an edge uniformly, putting residual $d-1$ on that edge and zero on the other three, and choosing a_0 uniformly. The remaining outcomes are

$$b_0 = a_0 - r_{00}, \qquad a_1 = b_0 - r_{10}, \qquad b_1 = a_1 - r_{11} \pmod d.$$

The fourth residual is consistent because the sum is $d-1$. This gives $4d$ deterministic strategies of weight $1/(4d)$. Every edge has residual probabilities $3/4, 1/4$ at $0, d-1$, and all marginals are uniform. Put

$$a_d = 4f_d(d-1) = \frac{2}{d^2 \sin^2(3\pi/(4d))}.$$

Lemma 6.1 (Uniform-marginal local component)**.** *For every $d \geq 2$,*

$$Q_d \geq a_d L_d, \qquad \frac{16}{45} < \frac{32}{9\pi^2} \leq a_d \leq \frac{8}{9}. \tag{6.2}$$

The coefficient a_d is maximal for this particular L_d. Furthermore,

$$v_{\mathrm{crit}}(Q_d) \geq \frac{1}{2 - a_d} > \frac{3}{5}. \tag{6.3}$$

Proof. With $x = \pi/(4d) \leq \pi/8$,

$$\frac{f_d(0)}{f_d(d-1)} = \left(\frac{\sin 3x}{\sin x}\right)^2 \geq (1+\sqrt{2})^2 > 3.$$

This gives entrywise domination. The remaining bounds in (6.2) use $2t/\pi \leq \sin t \leq t$ and $\pi^2 < 10$. The residual-$(d-1)$ entries make a_d tight for L_d.

For any no-signalling behaviour q_{rem} with uniform marginals,

$$(q_{\mathrm{rem}} + u_d)/2 \in \mathcal{C}_d. \tag{6.4}$$

A local model guesses Alice's input uniformly, samples her guessed output uniformly, and samples both Bob outputs conditionally according to q_{rem}. Alice's other output is independent and uniform. The guessed context gives q_{rem}, and the other gives u_d. Apply this to the normalized remainder of (6.2) to obtain (6.3). In particular, $Q_d = \frac{1}{3}L_d + \frac{2}{3}q_{\mathrm{rem}}$, with a uniform-marginal no-signalling remainder, gives a local model at visibility $3/5$. □

6.2. **Tensor measurements and noise separation.** Let $d = 2^m$, identify outcomes with bit strings, and use $\Phi_2^{\otimes m} = \Phi_{2^m}$. On each tensor factor measure

$$A_0 = Z, \qquad A_1 = X, \qquad B_0 = (Z+X)/\sqrt{2}, \qquad B_1 = (Z-X)/\sqrt{2}.$$

Each party uses the same input on all factors and retains every bit. These are two complete rank-one PVMs per party. Put

$$K_{xy}(a,b) = \#\{j : a_j \oplus b_j \neq xy\}, \qquad r = \frac{2-\sqrt{2}}{4}.$$

Equivalently, K_{xy} is the Hamming weight of $a \oplus b \oplus (xy)\mathbf{1}_m$, where $\mathbf{1}_m$ is the all-ones string. The behaviour is

$$T_m(a,b|x,y) = 2^{-m}(1-r)^{m-K_{xy}} r^{K_{xy}}. \tag{6.5}$$

For every deterministic local assignment, parity at each coordinate gives

$$\sum_{x,y} K_{xy} \geq m. \tag{6.6}$$

This applies to arbitrary dependence among the local output bits.

Theorem 1.3 (restated). *For d-outcome projective measurements on Φ_n with $I_d(q) > 2$, one has $v_{\mathrm{CGLMP}}(q) = 2/I_d(q) \geq 2/I_d^{\mathrm{F}}$, with equality precisely for standard amplifications. For every integer $m \geq 20000$,*

$$v_{\mathrm{crit}}(T_m) \leq \frac{17}{29} < \frac{3}{5} < v_{\mathrm{crit}}(Q_{2^m}).$$

Proof of Theorem 1.3. Since $I_d(u_d) = 0$,

$$I_d(vq + (1-v)u_d) = vI_d(q).$$

Theorem 1.1 therefore proves (1.5), including the equality characterization for every ambient dimension.

For $m \geq 20000$, define the Bell functional

$$G_m(p) = \frac{1}{4}\sum_{x,y}\left[\Pr_p(K_{xy} \leq 4m/25) - \Pr_p(K_{xy} \geq 13m/25)\right].$$

Its coefficients are $0, \pm 1/4$, and its local bound is $1/2$. Four low counts contradict (6.6); three low counts force the remaining count to be high.

Under T_m, $K \sim \mathrm{Bin}(m,r)$, where $r < 3/20$ and the variance is $m/8$. Each unwanted tail is bounded by

$$\frac{m/8}{(m/100)^2} = \frac{1250}{m} \leq \frac{1}{16}.$$

Thus $G_m(T_m) \geq 7/8$. Under u_d, $K \sim \mathrm{Bin}(m,1/2)$, and the upper tail at $13m/25$ is at most

$$\frac{m/4}{(m/50)^2} = \frac{625}{m} \leq \frac{1}{32}.$$

Hence $G_m(u_d) \geq -1/32$, and

$$G_m(vT_m + (1-v)u_d) \geq \frac{29v-1}{32}.$$

Locality requires $v \leq 17/29$. Together with Lemma 6.1, this proves (1.6). At $v = 3/5$, the tensor witness is at least

$$\frac{3}{5}\frac{7}{8} - \frac{2}{5}\frac{1}{32} = \frac{41}{80} > \frac{1}{2},$$

whereas Q_{2^m} has the local model above. □

The tensor probabilities lie in $\mathbb{Q}(\sqrt{2})$. Their binomial tails can be computed from $b_0 = (1-p)^m$ and $b_{k+1} = b_k(m-k)p/((k+1)(1-p))$.

6.3. **The universal statistical ceiling.**

Theorem 6.2 (Ceiling and closed form)**.** *For every no-signalling behaviour and every fixed input law,*

$$\mathcal{S}_\mu(q) \le [1 - \mathrm{LF}(q)]\Xi(\mu) \le \Xi(\mu). \tag{6.7}$$

If some $\mu_e = 0$, then $\Xi(\mu) = 0$. For full-support μ, there is a unique $\lambda > 0$ satisfying

$$\sum_e \min\{1, \mu_e/\lambda\} = 3,$$

and

$$\Xi(\mu) = \sum_{\mu_e < \lambda} \mu_e \log_2(\lambda/\mu_e). \tag{6.8}$$

Moreover, $\Xi(\mu) \le c_{\mathrm{KL}} := \log_2(4/3)$, with equality exactly for uniform inputs.

Proof. Delete an edge e from the four-cycle. The remaining three consistent pair distributions extend to a joint distribution on all four outputs by conditional gluing. Deleting 11, for example, gives

$$q(a_0, b_0|00)\, q(a_1|b_0, 10)\, q(b_1|a_0, 01).$$

Conditionals on zero-marginal events may be chosen arbitrarily. Let $L^{(e)}$ be the resulting local extension. For an optimizer s in (2.6), mix these extensions with weights $1 - s_e$. The local behaviour L_* satisfies $(L_*)_e \ge s_e q_e$, hence $D_\mu(q\|L_*) \le \Xi(\mu)$. Apply this comparison to the no-signalling remainder in $q = \alpha L + (1-\alpha)q_{\mathrm{rem}}$, then use joint convexity of relative entropy to obtain (6.7).

The minimum is unchanged by replacing $\sum_e s_e = 3$ with $\sum_e s_e \le 3$, since the objective decreases in each sampled coordinate. If an input is absent, choose $s_e = 1$ on all sampled contexts. For full support, the convex optimality conditions give $s_e = \min(1, \mu_e/\lambda)$, proving (6.8). To evaluate it, sort the four weights, test capped sets of sizes $k = 0, 1, 2$, and use

$$\lambda = \frac{\sum_{\text{uncapped}} \mu_e}{3 - k}.$$

Choose the set for which capped weights are at least λ and uncapped weights at most λ; ties contribute zero. Finally $s_e = 3/4$ gives $\Xi(\mu) \le c_{\mathrm{KL}}$. For nonuniform μ, a tangent perturbation at this interior point strictly decreases the objective. □

6.4. **Positive local content of quantum behaviours.**

Theorem 6.3 (Local-component bound)**.** *For arbitrary finite-dimensional states and POVMs in the $(2, 2, d)$ scenario,*

$$\mathrm{LF}(q) \ge \gamma_d := d - \sqrt{d^2 - 1} = \frac{1}{d + \sqrt{d^2 - 1}} > \frac{1}{2d}. \tag{6.9}$$

More generally, if Alice's two measurements have m_0, m_1 outcomes, the bound is

$$\frac{m_A - \sqrt{m_A^2 - 4}}{2}, \qquad m_A = m_0 + m_1.$$

This explicit dependence on the outcome counts continues the two-input positive-content results of [25, 26].

Proof. Purify the state and dilate the finite POVMs locally to PVMs. The dilation is the isometry $\xi \mapsto \sum_a \sqrt{F_a}\xi \otimes |a\rangle$, extended to a unitary separately for each setting with a common initial ancilla. We may therefore use a vector ψ, Alice projections A_a, C_c, and Bob projections B_b^y.

Set

$$u_{ab}^y = A_a B_b^y \psi, \qquad v_{cb}^y = C_c B_b^y \psi, \qquad p_{ab}^y = \left\|u_{ab}^y\right\|^2, \qquad r_{cb}^y = \left\|v_{cb}^y\right\|^2,$$
$$z_{acb}^y = \langle u_{ab}^y, v_{cb}^y\rangle, \qquad k_{ac} = \langle A_a\psi, C_c\psi\rangle, \qquad w_{ac} = q_A(a|0) + q_A(c|1).$$

Then

$$\sum_b z_{acb}^y = k_{ac}, \qquad \sum_{a,c} k_{ac} = 1, \qquad \sum_{a,c} w_{ac} = m_A.$$

Define, with $0/0 = 0$,

$$t_{acb}^y = \frac{|z_{acb}^y|^2}{\max\{p_{ab}^y, r_{cb}^y\}}.$$

At fixed y, b, the nonzero vectors $v_{cb}^y/\sqrt{r_{cb}^y}$ are orthonormal in c. Bessel's inequality and its counterpart in a give

$$\sum_c t_{acb}^y \le p_{ab}^y, \qquad \sum_a t_{acb}^y \le r_{cb}^y. \tag{6.10}$$

Fix a, c, abbreviate $w = w_{ac}$, $k = |k_{ac}|$, and put

$$h_{ac} = \frac{w - \sqrt{w^2 - 4k^2}}{2},$$

with value zero if $w = 0$. Cauchy–Schwarz gives $k \le w/2$. For $T_y = \sum_b t_{acb}^y$ and $z_y = \sum_b \min(p_{ab}^y, r_{cb}^y)$, one has $z_y \le w/2$ and

$$k \le \sum_b \sqrt{p_{ab}^y r_{cb}^y} \le \sqrt{z_y(w - z_y)}.$$

Thus $z_y \ge h_{ac}$. Weighted Cauchy–Schwarz gives, for $k > 0$,

$$T_y \ge \frac{k^2}{w - z_y} \ge \frac{k^2}{w - h_{ac}} = h_{ac};$$

the zero case follows directly.

Assign the deterministic strategy (a, c, b_0, b_1) weight

$$w^{\mathrm{loc}}(a, c, b_0, b_1) = h_{ac} \frac{t_{acb_0}^0}{T_0} \frac{t_{acb_1}^1}{T_1}, \tag{6.11}$$

with value zero for $h_{ac} = 0$. Otherwise the denominators are positive. By (6.10) and $h_{ac} \le T_y$, the weights (6.11) define a subnormalized local behaviour entrywise below q, of mass $\alpha = \sum_{a,c} h_{ac}$.

The function $\phi(t) = (1 - \sqrt{1 - 4t^2})/2$ is convex and increasing on $[0, 1/2]$. Weighted Jensen, omitting zero weights, yields

$$\alpha = \sum_{a,c} w_{ac}\phi(|k_{ac}|/w_{ac}) \ge m_A \phi\left(\frac{\sum_{a,c}|k_{ac}|}{m_A}\right) \ge m_A\phi(1/m_A),$$

where $\sum |k_{ac}| \ge 1$ was used in the last step. This proves the general bound, and $m_A = 2d$ gives (6.9). □

Theorems 6.2 and 6.3 imply, uniformly in the state dimension,

$$\mathcal{S}_\mu(q) \le (1 - \gamma_d)\Xi(\mu) \tag{6.12}$$

for every finite-outcome quantum behaviour. For full-support μ, $\Xi(\mu) > 0$, giving a strict gap. If a context is absent, the sampled graph is a forest, and every no-signalling behaviour has zero strength on the sampled data.

6.5. **The supremum and an explicit separation.**

Theorem 1.4 (restated). *For every fixed input law independent of the hidden variable,*

$$\sup_{\substack{d\geq 2 \\ \text{two rank-one PVMs per party on } \Phi_d}} \mathcal{S}_\mu(q) = \Xi(\mu).$$

The same supremum holds over arbitrary finite-dimensional quantum states and finite-outcome POVMs. For full-support μ, every finite-outcome quantum behaviour has strength strictly below $\Xi(\mu)$.

Proof of Theorem 1.4. Use T_m from (6.5) and the strict events $\mathcal{A}_{xy} = \{K_{xy} < m/4\}$, with common quantum probability

$$t_m = \Pr\{\mathrm{Bin}(m, r) < m/4\}.$$

By (6.6), at most three of the four events hold on a deterministic local assignment. Thus local event probabilities obey $\sum_e z_e \leq 3$. Data processing, retaining the context and both the event and its complement, gives

$$\mathcal{S}_\mu(T_m) \geq \min_{\substack{0\leq z_e\leq 1 \\ \sum_e z_e\leq 3}} \sum_e \mu_e D_{\mathrm{bin}}(t_m \| z_e) \geq t_m \Xi(\mu) - h_2(t_m).$$

The final inequality drops the nonnegative terms $-(1-t_m)\log_2(1-z_e)$.

A centered Bernoulli log moment-generating function has second derivative at most $1/4$. Integrating twice and optimizing the Chernoff bound gives

$$1 - t_m \leq e^{-\kappa m}, \qquad \kappa = 2(1/4 - r)^2 = \frac{3 - 2\sqrt{2}}{8} > \frac{1}{50},$$

where the last comparison follows from $(71/50)^2 > 2$. Using $h_2(\epsilon) \leq \epsilon \log_2(e/\epsilon)$, we obtain

$$0 \leq \Xi(\mu) - \mathcal{S}_\mu(T_m) \leq e^{-\kappa m}\left[\Xi(\mu) + \frac{\kappa m + 1}{\ln 2}\right]. \tag{6.13}$$

This tends to zero. Theorem 6.2 supplies the universal upper bound, and (6.12) supplies the finite-outcome quantum gap, proving (1.7). □

The binary PR behaviour attains $\Xi(\mu)$ in $\mathcal{P}_2$: its four parity tests succeed with certainty, while local success probabilities satisfy $\sum_e z_e \leq 3$. For uniform inputs this gives the value computed in [24].

From (6.2) and (6.7),

$$\mathcal{S}_\mu(Q_d) \leq (1 - a_d)\Xi(\mu), \qquad \sup_\mu \mathcal{S}_\mu(Q_d) < \frac{29}{45}\log_2\frac{4}{3}. \tag{6.14}$$

For $m \geq 200$, $\kappa m > 4$. The function $e^{-x}[c_{\mathrm{KL}} + (x+1)/\ln 2]$ decreases for $x \geq 0$, and

$$\ln(4/3) = 2\,\mathrm{artanh}(1/7) > \frac{2}{7}, \qquad e^4 > \sum_{j=0}^{9} \frac{4^j}{j!} > 54.$$

Equation (6.13) therefore gives

$$\frac{c_{\mathrm{KL}} - \mathcal{S}_{\mathrm{unif}}(T_m)}{c_{\mathrm{KL}}} < \frac{1}{54}\left(1 + \frac{5}{2/7}\right) = \frac{37}{108}.$$

Since $71/108 = 29/45 + 7/540$, combining this with (6.14) proves, for every $m \geq 200$,

$$\mathcal{S}_{\mathrm{unif}}(T_m) > \frac{71}{108}\log_2\frac{4}{3} > \frac{29}{45}\log_2\frac{4}{3} > \sup_{\mu}\mathcal{S}_\mu(Q_{2^m}). \tag{6.15}$$

6.6. **Statistical strength near CGLMP optimality.** For behaviours in the same finite scenario,

$$|\,\mathrm{LF}(p) - \mathrm{LF}(q)| \leq \sum_{x,y}\mathrm{TV}(p_{xy}, q_{xy}). \tag{6.16}$$

To prove this, write a sublocal component $F \leq p$ with deterministic weights w_v. For each event e, set $t_e = (F_e - q_e)_+/F_e$, with value zero if $F_e = 0$, and replace

$$w_v \longmapsto w_v \prod_{\text{contexts } c}(1 - t_{c,v(c)}).$$

The resulting sublocal component lies below q. The union bound controls its lost mass by

$$\sum_e (F_e - q_e)_+ \leq \sum_c \mathrm{TV}(p_c, q_c).$$

Interchanging p, q proves (6.16).

Let P be d-outcome PVMs on Φ_{dk}, and choose a standard reference Q realizing (2.3). Write $\varepsilon_{\mathrm{meas}} = \varepsilon_{\mathrm{meas}}(P, Q)$. For each context, the identity $\sqrt{\tau(PR)} = \|PR\|_2$, completeness, and the triangle inequality give

$$H(q_{xy}, Q_{d,xy}) \leq \sqrt{e_{2x}} + \sqrt{e_{1+2y}}, \qquad e_t = \sum_a \left\|P_a^t - Q_a^t\right\|_2^2.$$

Since $\mathrm{TV} \leq H$, $\sum_{x,y}\mathrm{TV}(q_{xy}, Q_{d,xy}) \leq 4\sqrt{\varepsilon_{\mathrm{meas}}}$. Theorem 1.2 and (6.16), with $\widehat{K}_d = 786432d^6$, imply

$$\mathrm{LF}(q) \geq \left(a_d - 4\sqrt{\widehat{K}_d\Delta}\right)_+.$$

Together with (6.9),

$$\mathcal{S}_\mu(q) \leq \left[1 - \max\left\{\gamma_d, \left(a_d - 4\sqrt{\widehat{K}_d\Delta}\right)_+\right\}\right]\Xi(\mu). \tag{6.17}$$

The uniform-noise construction (6.4) uses the uniform-marginal component L_d; the local-fraction estimates use entrywise domination for general marginals.

7. Discussion

For projective measurements on maximally entangled states, maximal CGLMP violation determines the Fourier-phase measurements and their multiplicity. The quantitative bounds also detect the arithmetic obstruction to equality and give an optimal deficit of order $1/n$ along fixed nonzero residue classes. The operational comparisons distinguish optimization of the CGLMP witness from global noise resistance and statistical strength. The function $\Xi(\mu)$ determines the dimension-unrestricted statistical ceiling, while positive local content gives a finite-outcome gap and (6.17) relates that gap to projective CGLMP rigidity.

Several optimization problems remain open. For d-outcome POVMs on a fixed Φ_n, determine the maximum CGLMP value and classify the maximizing measurements. For optimization over states, determine the sharp value and all equality cases at general d, extending the low-dimensional analytic results [10]. When $n \bmod d > 0$, determine the projective optimum and the structure of its maximizers beyond the order estimate (1.4).

For a prescribed finite outcome count and input law, determine the statistical-strength optimum and its optimizing state and measurements. A related question is to determine the sharp outcome-dependent local-fraction bound, including its dependence on unequal outcome counts. These problems would sharpen both the finite-dimensional ceiling gap and the relation between Bell-value optimality and statistical evidence.

Code availability

Scripts executing the rational interval and Bernstein-coefficient calculations, and evaluating the integer facet minor, are available from the authors. They use the data and procedures described in Appendices A–D.

Appendix A. The Bancal–Gisin–Pironio four-outcome facet

Bancal, Gisin, and Pironio [18, Appendix B] found the inequality $S^2_{(2,2,4)}$ as a genuinely four-outcome facet. We give its integer rank certificate, a rational no-signalling point, and the outcome-fusion argument.

Use each party's six independent marginals, for example

$$p_A = (p_A(0|0), p_A(1|0), p_A(2|0), p_A(0|1), p_A(1|1), p_A(2|1)),$$

and the 36 joint probabilities with $a, b < 3$. The affine dimension is 48. The inequality is

$$F(p) = \ell \cdot p_A + \ell \cdot p_B + \sum_{x,y=0}^{1} \sum_{a,b=0}^{2} H_{3x+a,\,3y+b}\, p(a,b|x,y) \le 0, \tag{A.1}$$

where

$$\ell = (-1,-1,0,-1,0,0), \qquad H = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}.$$

Proposition A.1. *Inequality* (A.1) *defines a facet of $\mathcal{C}_4$ meeting the relative interior of $\mathcal{P}_4$. Its class is distinct from CGLMP$_4$ and from every outcome-fusion lifting of a lower-outcome inequality. There is a rational no-signalling behaviour p^+ with $F(p^+) = 1/144$. Four is the smallest outcome count for such a facet.*

Proof. For $s = (a_0, a_1, b_0, b_1)$, define

$$A(s) = (\mathbf{1}_{a_0=0}, \mathbf{1}_{a_0=1}, \mathbf{1}_{a_0=2}, \mathbf{1}_{a_1=0}, \mathbf{1}_{a_1=1}, \mathbf{1}_{a_1=2}),$$

and define $B(s)$ analogously. Then

$$F(s) = \ell \cdot A(s) + \ell \cdot B(s) + A(s)^T H B(s).$$

Evaluation on the 4^4 strategies gives Table 1, proving local validity. Let $\mathcal{S} = \{s : F(s) = 0\}$, and augment the coordinates:

$$z(s) = \big(1, A(s), B(s), (A_i(s)B_j(s))_{i,j=0}^{5}\big), \qquad \iota(s) = 64a_0 + 16a_1 + 4b_0 + b_1,$$

with the products in row-major order. Select the rows with the ordered identifiers in Table 2. Numbering columns from zero and deleting column 47 gives a 48×48 matrix $Z_{\min}$ with

$$\det Z_{\min} = -1. \tag{A.2}$$

Thus the saturation face has affine rank 47.

The average

$$\bar{p} = \frac{1}{72} \sum_{s \in \mathcal{S}} q_s$$

has minimum entry $1/72$, obtained by counting occurrences of each event among the saturators. Since probability nonnegativity defines $\mathcal{P}_4$ in its affine hull, $\bar{p}$ is a relative-interior point. Modify only context 00, adding $1/144$ at $(0,1),(3,3)$ and subtracting $1/144$ at $(0,3),(3,1)$. The resulting p^+ preserves all row and column sums, remains nonnegative, and satisfies $F(p^+) = 1/144$.

A deterministic CGLMP_d saturator is specified by an anchor outcome and four nonnegative residuals summing to $d-1$. Its saturation count is $d\binom{d+2}{3}$. At $d = 4$, the counts 80 and 72 distinguish the two classes.

Two outcomes are twins when replacing one by the other preserves the functional for every assignment of the other three outcomes. Every nontrivial fusion lifting has twins. Here the four extended coefficient rows, including the zero row for output 3, together with their marginal coefficients, are distinct within each setting. The analogous column profiles are also distinct. Taking the other party's unused outputs to be 3 isolates any differing coefficient. This distinguishes every outcome pair and separates the facet from all nontrivial fusion liftings.

The complete two-outcome classification is due to Fine [19]; the three-outcome list uses the classifications and facet results of [20, 4]. Their counts in Table 3 show that $d = 4$ is the first outcome count with the asserted properties. □

TABLE 1. Deterministic values of (A.1).

$F(s)$	0	−1	−2
Number of strategies	72	144	40

TABLE 2. Ordered saturator identifiers for the minor (A.2).

4	6	8	20	21	22	23	34	38	42	46	54
58	62	64	65	72	81	85	89	93	96	97	98
99	113	121	125	128	132	136	140	149	151	156	162
163	172	188	191	200	201	202	203	213	215	221	226

TABLE 3. Complete facet counts for two and three outcomes.

d	Positivity	Fused CHSH	CGLMP_3
2	16	8	0
3	36	648	432

For general d, the independent coordinates are each party's $2(d-1)$ marginals and the $4(d-1)^2$ joint probabilities with $a, b < d-1$. For example,

$$p(a, d-1|x,y) = p_A(a|x) - \sum_{b<d-1} p(a,b|x,y),$$

with the analogous Bob formula and normalization determining the remaining entries. Mixed differences of deterministic vertices show full affine span.

The small-outcome lists can also be obtained by the rational double-description method [28]. With $L = 4d(d-1)+1$, start from the columns of Z^{-1}, where Z is a greedily selected set of L

independent augmented rows. Insert the other rows lexicographically. For a new row z, retain rays with $zr \geq 0$, and join a positive and negative ray by $(zr_+)r_- - (zr_-)r_+$ when their common active rows have rank $L-2$. This rank condition identifies a two-dimensional face. Primitive-integer normalization removes duplicate rays. Comparing final value vectors, up to positive proportionality, with positivity, relabelled CGLMP functionals, and CGLMP_2 composed with output surjections gives Table 3. For four outcomes, the larger list of Jesus and Zambrini Cruzeiro contains 34 classes and 11 665 992 facets [21]; adjacency sampling provides a complementary enumeration method [22].

Appendix B. The circulation sum of squares

Fix $i < j < k$, and write

$$\begin{aligned} u_\ell &= E_{i\ell}E_{j,\ell+2}, \quad v_\ell = E_{j\ell}E_{k,\ell+2}, \quad z_\ell = E_{i\ell}E_{k,\ell+2}, \\ a_\ell &= E_{i\ell}E_{j,\ell+1}, \quad b_\ell = E_{j\ell}E_{k,\ell+1}, \quad c_\ell = E_{i\ell}E_{k,\ell+1}. \end{aligned}$$

Let P be the sum of squared norms of a_ℓ, b_ℓ, c_ℓ, and O the corresponding sum for u_ℓ, v_ℓ, z_ℓ. Then $P + O = p_{ij} + p_{jk} + p_{ik}$. Use the triples in Table 4 for each $\ell \in \mathbb{Z}_4$.

Table 4. Terms in the circulation identity.

$X_{\ell\nu}$	$Y_{\ell\nu}$	ϵ_ν
z_ℓ	a_ℓ	1
$z_\ell^\dagger$	$b_{\ell+1}^\dagger$	1
$v_{\ell+1}$	$a_\ell^\dagger$	1
u_ℓ	c_ℓ	-1
$u_\ell^\dagger$	$b_{\ell+2}$	1
$v_{\ell+3}^\dagger$	$c_\ell^\dagger$	-1

Projection completeness gives

$$\begin{aligned} C_{ijk} = &-2\sum_\ell \|z_\ell\|_2^2 + 2\Re\sum_\ell \left[\langle u_\ell, z_\ell\rangle + \langle v_\ell^\dagger, z_\ell^\dagger\rangle\right] \\ &+ 2\Re\sum_{\ell,\nu} \epsilon_\nu \langle X_{\ell\nu}, Y_{\ell\nu}\rangle. \end{aligned}$$

For $\sigma = \pm 1$,

$$\begin{aligned} 2P + 5O - \sigma C_{ijk} = &\sum_{\ell,\nu} \|Y_{\ell\nu} - \sigma\epsilon_\nu X_{\ell\nu}\|_2^2 \\ &+ 3\sum_\ell \left\|u_\ell - \tfrac{\sigma}{3} z_\ell\right\|_2^2 + 3\sum_\ell \left\|v_\ell^\dagger - \tfrac{\sigma}{3} z_\ell^\dagger\right\|_2^2 \\ &+ \left(\frac{7}{3} + 2\sigma\right)\sum_\ell \|z_\ell\|_2^2 \geq 0. \end{aligned} \tag{B.1}$$

Thus $|C_{ijk}| \leq 2P + 5O \leq 5(P + O)$.

Both identities follow by expanding $E_{j3} = I - E_{j0} - E_{j1} - E_{j2}$, reducing adjacent same-site products by orthogonality, and comparing coefficients of cyclic words. Choose the lexicographically least cyclic rotation of each word, and expand each real part into the half-sum of a word and its adjoint, keeping those coefficients separate.

Appendix C. Scalar coefficients and the alternative estimate

C.1. **Coefficient and interval data.** The coefficients of $r_{\mathrm{reg}}(y) = \csc(\pi y/2) - 2/(\pi y)$ are

$$r_n = \frac{2(2^{2n-1}-1)|B_{2n}|}{(2n)!}\left(\frac{\pi}{2}\right)^{2n-1}, \qquad 0 < r_n < 3\,4^{-n}, \qquad r_{\mathrm{reg}}(y) = \sum_{n\geq 1} r_n y^{2n-1}. \tag{C.1}$$

For $y = 1 - h/d$, let $m_k = \frac{d}{d+1}(d-h+1)_k/(d+2)_k$. The error in (4.6) is

$$e_{d,h} = \sum_{n\geq 1} r_n \left[y^k - \left(1 + \frac{(k+1)(k+2)}{2d}\right) m_k + \frac{k(k+1)}{2d} m_{k-1} \right], \qquad k = 2n-1. \tag{C.2}$$

The bracket is bounded in absolute value by $2 + 4n^2/(d+1)$. Truncation after 32 terms has error at most

$$T_d = 3\sum_{n=33}^{\infty} 4^{-n}\left(2 + \frac{4n^2}{d+1}\right). \tag{C.3}$$

For $q = 1/4$, the tail is evaluated using

$$\sum_{n\geq J} q^n = \frac{q^J}{1-q}, \qquad \sum_{n\geq J} n^2 q^n = q^J\left[\frac{J^2}{1-q} + \frac{2Jq}{(1-q)^2} + \frac{q(1+q)}{(1-q)^3}\right].$$

For the coefficient enclosures, put

$$A(k,N) = \sum_{j=0}^{N-1} \frac{(-1)^j}{(2j+1)k^{2j+1}}.$$

Machin's identity gives the interval

$$\pi \in 16[A(5,128), A(5,129)] - 4[A(239,128), A(239,129)].$$

Generate $B_0 = 1$ and $B_m = -(m+1)^{-1}\sum_{j=0}^{m-1}\binom{m+1}{j}B_j$ through $m = 64$, and substitute the interval for π into (C.1). For (C.2), use $m_0 = d/(d+1)$ and $m_k = m_{k-1}(d-h+k)/(d+k+1)$. Endpoint inclusion in these rational operations, followed by $[-T_d, T_d]$, encloses each $e_{d,h}$. For interval triangle loads in (4.14), use $\mathrm{amag}([l,u]) = \max(|l|,|u|)$. The parameters and resulting lower bound are collected in Table 5.

Table 5. Data for the finite interval calculation. The margin uses lower endpoints for $e_{d,h}$ and upper bounds for $\ell_{d,h}$.

Quantity	Data
Outcome counts	$2 \leq d \leq 65$
Separations	$1 \leq h < d$
Cosecant terms	$1 \leq n \leq 32$
Moment indices	$0 \leq k \leq 63$
Bernoulli indices	$0 \leq m \leq 64$
Arctangent partial sums	128, 129 terms
Tail start in (C.3)	$J = 33$
Lower bound for the normalized margin	$1935/10000 > 1/6$

Consequently, for every $2 \leq d \leq 65$,

$$(d+1)\left[\frac{1}{3(d+1)} + \min_h e_{d,h} - \frac{5}{2d}\max_h \ell_{d,h}\right] > \frac{1}{6}. \tag{C.4}$$

This is the finite-range input to (4.16).

C.2. **The beta–Stein estimate.** Put $\varphi(y) = y(1-y)r_{\mathrm{reg}}(y)$ and $Z \sim \mathrm{Beta}(dy, d(1-y))$. For $y = 1 - h/d$,

$$e_{d,h} = \frac{\varphi(y) - \mathbb{E}\varphi(Z) + \frac{1}{2}\mathbb{E}[(Z-y)\varphi'(Z)]}{y(1-y)}. \tag{C.5}$$

Indeed, the pole cancels because $B_d((1-x)^{-1})(h) = d/(d-h)$. Size bias replaces $B_d g$ by $\mathbb{E}[Z(1-Z)g(Z)]/[y(1-y)]$, and

$$\mathbb{E}[(Z-y)g(Z)] = d^{-1}\mathbb{E}[Z(1-Z)g'(Z)].$$

These identities give (C.5).

For $k = 2n-1$, the third and fourth derivatives of $y^{k+1} - y^{k+2}$ are bounded by $3k(k+1)$ and $4k(k-1)(k+1)$. Summing the positive series (C.1) and using

$$r''_{\mathrm{reg}}(1) + 2r'_{\mathrm{reg}}(1) = \pi^2/4, \qquad r'''_{\mathrm{reg}}(1) + 3r''_{\mathrm{reg}}(1) = 3\pi^2/4$$

gives $\|\varphi'''\|_\infty \le 3\pi^2/4$ and $\|\varphi^{(4)}\|_\infty \le 3\pi^2$. The trapezoidal remainder is

$$\varphi(y) - \varphi(y+\xi) + \frac{\xi}{2}[\varphi'(y+\xi) + \varphi'(y)] = \frac{\xi^3}{12}\varphi'''(y) + \epsilon_\xi,$$
$$|\epsilon_\xi| \le \frac{\|\varphi^{(4)}\|_\infty}{24}|\xi|^4.$$

For $\xi = Z - y$,

$$\mathbb{E}\xi^3 = \frac{2y(1-y)(1-2y)}{(d+1)(d+2)}, \qquad \mathbb{E}\xi^4 = \frac{3y(1-y)[2(1-2y)^2 + y(1-y)(d+2)]}{(d+1)(d+2)(d+3)}.$$

Thus, with $q = |1-2y|$,

$$|e_{d,h}| \le \frac{\pi^2}{8(d+1)(d+2)}\left[q + \frac{3(d+2)}{4(d+3)} - \frac{3(d-6)}{4(d+3)}q^2\right].$$

For $d \ge 66$, the bracket is at most

$$\frac{3(d+2)}{4(d+3)} + \frac{d+3}{3(d-6)} < \frac{9}{8},$$

the difference from $9/8$ being $(d^2 - 57d - 342)/(24(d-6)(d+3)) > 0$. Using $\pi^2 < 10$, we obtain

$$b_d \le \frac{45}{32(d+1)(d+2)}. \tag{C.6}$$

To derive (4.16), combine (C.6), (4.15), and $L_s \ge 2\mathfrak{F}_d/[3(d+1)]$ with (4.8). For $2 \le d \le 65$, instead use (4.13), (4.12), and (C.4):

$$L_{e+s} \ge 2\left[\frac{1}{3(d+1)} + \min_h e_{d,h} - \frac{5}{2d}\max_h \ell_{d,h}\right]\mathfrak{F}_d \ge \frac{\mathfrak{F}_d}{3(d+1)}.$$

Appendix D. Polynomial positivity on the continuous domain

For $\theta \ge 0$, define

$$Q_{k,\theta}(y) = \prod_{j=1}^{k} \frac{y + j\theta}{1 + (j+1)\theta}, \qquad Q_{0,\theta} = 1,$$

and

$$\mathcal{E}_{k,\theta}(y) = \frac{(1+\theta)y^k - Q_{k,\theta}(y)}{\theta} - \frac{(k+1)(k+2)}{2}Q_{k,\theta}(y) + \frac{k(k+1)}{2}Q_{k-1,\theta}(y),$$

with the removable value at zero filled continuously. Put $\mathcal{E}_\theta(y) = \sum_{n\ge1} r_n \mathcal{E}_{2n-1,\theta}(y)$. Formula (C.2) gives

$$\mathcal{E}_{1/d}(1-h/d) = (d+1)e_{d,h}. \tag{D.1}$$

The required scalar bound is

$$\begin{aligned} &\frac{1}{3} + \mathcal{E}_\theta(y) - \frac{5}{2}|\Omega_\theta(x_1,x_2,x_3)| \ge \frac{1}{6}, \\ &\Omega_\theta = (1-2\theta)\sum_{j=1}^{3}[\mathcal{E}_\theta(1-x_j) - \mathcal{E}_\theta(x_j)], \end{aligned} \tag{D.2}$$

for $0 < \theta \le 1/3$, $\theta \le y \le 1-\theta$, $x_j \ge \theta$, and $\sum_j x_j = 1$.

D.1. **Truncation data.** Let $Q = 10^{12}$, and take the integers in Table 6. The intervals from Appendix C give $m_n/Q < r_n < (m_n+1)/Q$. The partial-fraction expansion of cosecant yields

$$r_n = \frac{4}{\pi}\,4^{-n}\eta(2n) < \frac{4}{3}\,4^{-n}, \qquad \eta(2n) = \sum_{j\ge1}(-1)^{j-1}j^{-2n} \in (0,1).$$

Each factor $q_j = (y+j\theta)/(1+(j+1)\theta)$ belongs to $[0,1]$ for $0 \le y \le 1$, with $|q_j - y| \le j\theta$. Hence $|\mathcal{E}_{k,\theta}(y)| \le 1 + (k+1)^2$. Replacing r_n by m_n/Q for $n \le 10$ and omitting the remaining terms has uniform error at most

$$\varepsilon_{\mathrm{sc}} = \frac{1550}{10^{12}} + \frac{4}{3}\sum_{n=11}^{\infty} 4^{-n}(1+4n^2) = \frac{45400712033}{207360000000000} < \varepsilon_0 := \frac{9}{40000}. \tag{D.3}$$

TABLE 6. Integers specifying $m_n/Q < r_n < (m_n+1)/Q$, with $Q = 10^{12}$.

n	m_n	n	m_n
1	261799387799	6	310774174
2	75362478042	7	77707647
3	19606915981	8	19427797
4	4954856459	9	4857004
5	1242203718	10	1214254

D.2. **The polynomials and their Bernstein bounds.** Let

$$D_*(\theta) = \prod_{j=2}^{20}(1+j\theta), \qquad D_k = \prod_{j=2}^{k+1}(1+j\theta), \qquad P_k = \prod_{j=1}^{k}(y+j\theta),$$

with empty products one, and set $a_k = (k+1)(k+2)/2$, $b_k = k(k+1)/2$. Define

$$\begin{aligned} N(\theta,y) &= \sum_{\substack{1\le n\le 10\\ k=2n-1}} m_n \frac{D_*}{D_k}\frac{\mathcal{B}_k(\theta,y)}{\theta}, \\ \mathcal{B}_k(\theta,y) &= (1+\theta)y^k D_k - (1+a_k\theta)P_k + b_k\theta P_{k-1}(1+(k+1)\theta). \end{aligned}$$

Every $\mathcal{B}_k$ is divisible by θ^2. Thus N is an integer polynomial of bidegree at most $(19,19)$, and the truncated function is $N/(QD_*)$.

Parameterize the domains by

$$Y = \theta + (1-2\theta)u, \qquad X_j = \theta + (1-3\theta)\alpha_j, \qquad 0 \le u \le 1, \quad \alpha_j \ge 0, \quad \sum_j \alpha_j = 1,$$

and put

$$\mathcal{N}_{\triangle} = (1-2\theta)\sum_{j=1}^{3}[N(\theta, 1-X_j) - N(\theta, X_j)].$$

The polynomial inequalities are

$$100N(\theta, Y) + 3QD_*(\theta) > 0, \qquad 0 \le \theta \le 1/2, \tag{D.4}$$

and

$$(19568 + 810\theta)QD_* + 120000N(\theta, Y) \pm 300000\mathcal{N}_{\triangle} > 0, \qquad 0 \le \theta \le 1/3. \tag{D.5}$$

The degree-n interval Bernstein coefficient of θ^p on $[0, 1/b]$, at index i, is $b^{-p}\binom{i}{p}/\binom{n}{p}$. For u^q on $[0,1]$, use $\binom{j}{q}/\binom{m}{q}$. On the degree-$m$ simplex, the coefficient of $\alpha_1^{q_1}\alpha_2^{q_2}\alpha_3^{q_3}$ at $\gamma_1+\gamma_2+\gamma_3 = m$ is

$$\frac{\prod_{\ell=1}^{3}\gamma_\ell(\gamma_\ell-1)\cdots(\gamma_\ell-q_\ell+1)}{m(m-1)\cdots(m-q_1-q_2-q_3+1)}.$$

Empty ratios equal one and impossible indices give zero. Linearity and products give the coefficients on product domains.

For the rectangle, let $A_{ij}^{(2)}$ and $d_i^{(2)}$ be the coefficients of $N(\theta, Y)$ and D_* at degrees $(38, 19)$ and 38, on $\theta \in [0, 1/2]$. For the joint domain, let $A_{ij}^{(3)}, B_{i\gamma}, d_i, t_i$ be the coefficients of $N(\theta, Y), \mathcal{N}_{\triangle}, D_*, \theta D_*$ at the degrees in Table 7. Put

$$L_i = Q(19568d_i + 810t_i) + 120000\min_j A_{ij}^{(3)} - 300000\max_\gamma |B_{i\gamma}|.$$

The coefficient inequalities in Table 7, obtained by the displayed rational conversion formulas, imply (D.4) and both signs of (D.5). For the joint domain the normalized lower margin is greater than $1/6000$.

TABLE 7. Bernstein data. The simplex degree refers to $\alpha_1+\alpha_2+\alpha_3 = 1$.

Domain	Degrees and indices	Coefficient lower bounds
Rectangle: $0 \le \theta \le 1/2$	$\deg_\theta = 38$, $\deg_u = 19$; $0 \le i \le 38$, $0 \le j \le 19$	$100A_{ij}^{(2)} + 3Qd_i^{(2)} > 0$ (780 coefficients)
Joint: $0 \le \theta \le 1/3$	$\deg_\theta = 39$, $\deg_u = 19$, $\deg_{\text{simplex}} = 19$; $0 \le i \le 39$	$L_i > 20Qd_i$ (40 joint bounds)

Dividing (D.5) by $120000QD_*$ gives

$$\frac{1}{6} - \varepsilon_0(16 - 30\theta) + \mathcal{E}_\theta^{[10]}(Y) \pm \frac{5}{2}\Omega_\theta^{[10]}.$$

The superscript denotes the ten-term replacement from Table 6. By (D.3), the omitted terms have total magnitude at most $\varepsilon_0 + 15(1-2\theta)\varepsilon_0$. This proves (D.2). Likewise, (D.4) gives

$$\mathcal{E}_\theta(y) > -\frac{3}{100} - \varepsilon_0, \qquad 0 \le \theta \le 1/2, \quad \theta \le y \le 1-\theta. \tag{D.6}$$

REFERENCES

[1] A. Salavrakos, R. Augusiak, J. Tura, P. Wittek, A. Acín, and S. Pironio. Bell inequalities tailored to maximally entangled states. *Phys. Rev. Lett.*, 119:040402, 2017. doi:10.1103/PhysRevLett.119.040402; arXiv:1607.04578.

[2] S. Sarkar, D. Saha, J. Kaniewski, and R. Augusiak. Self-testing quantum systems of arbitrary local dimension with minimal number of measurements. *npj Quantum Inf.*, 7:151, 2021. doi:10.1038/s41534-021-00490-3; arXiv:1909.12722.

[3] D. Collins, N. Gisin, N. Linden, S. Massar, and S. Popescu. Bell inequalities for arbitrarily high-dimensional systems. *Phys. Rev. Lett.*, 88:040404, 2002. doi:10.1103/PhysRevLett.88.040404; arXiv:quant-ph/0106024.

[4] Ll. Masanes. Tight Bell inequality for d-outcome measurements correlations. *Quantum Inf. Comput.*, 3(4):345–358, 2003. doi:10.26421/QIC3.4-4; arXiv:quant-ph/0210073.

[5] T. Durt, D. Kaszlikowski, and M. Żukowski. Violations of local realism with quantum systems described by N-dimensional Hilbert spaces up to $N = 16$. *Phys. Rev. A*, 64:024101, 2001. doi:10.1103/PhysRevA.64.024101; arXiv:quant-ph/0101084.

[6] S. Zohren and R. D. Gill. Maximal violation of the CGLMP inequality for infinite dimensional states. *Phys. Rev. Lett.*, 100:120406, 2008. doi:10.1103/PhysRevLett.100.120406; arXiv:quant-ph/0612020.

[7] B. Lang, T. Vértesi, and M. Navascués. Closed sets of correlations: answers from the zoo. *J. Phys. A: Math. Theor.*, 47:424029, 2014. doi:10.1088/1751-8113/47/42/424029; arXiv:1402.2850.

[8] A. Acín, T. Durt, N. Gisin, and J. I. Latorre. Quantum nonlocality in two three-level systems. *Phys. Rev. A*, 65:052325, 2002. doi:10.1103/PhysRevA.65.052325; arXiv:quant-ph/0111143.

[9] M. Navascués, S. Pironio, and A. Acín. Bounding the set of quantum correlations. *Phys. Rev. Lett.*, 98:010401, 2007. doi:10.1103/PhysRevLett.98.010401; arXiv:quant-ph/0607119.

[10] M. Ioannou and D. Rosset. Noncommutative polynomial optimization under symmetry. arXiv preprint, 2021. arXiv:2112.10803.

[11] T. H. Yang, T. Vértesi, J.-D. Bancal, V. Scarani, and M. Navascués. Robust and versatile black-box certification of quantum devices. *Phys. Rev. Lett.*, 113:040401, 2014. doi:10.1103/PhysRevLett.113.040401; arXiv:1406.7127.

[12] J. Kaniewski, I. Šupić, J. Tura, F. Baccari, A. Salavrakos, and R. Augusiak. Maximal nonlocality from maximal entanglement and mutually unbiased bases, and self-testing of two-qutrit quantum systems. *Quantum*, 3:198, 2019. doi:10.22331/q-2019-10-24-198; arXiv:1807.03332.

[13] M. Farkas and J. Kaniewski. Self-testing mutually unbiased bases in the prepare-and-measure scenario. *Phys. Rev. A*, 99:032316, 2019. doi:10.1103/PhysRevA.99.032316; arXiv:1803.00363.

[14] W. van Dam, R. D. Gill, and P. D. Grünwald. The statistical strength of nonlocality proofs. *IEEE Trans. Inform. Theory*, 51(8):2812–2835, 2005. doi:10.1109/TIT.2005.851738; arXiv:quant-ph/0307125.

[15] A. Acín, R. Gill, and N. Gisin. Optimal Bell tests do not require maximally entangled states. *Phys. Rev. Lett.*, 95:210402, 2005. doi:10.1103/PhysRevLett.95.210402; arXiv:quant-ph/0506225.

[16] K. Baek, J. Ryu, and J. Lee. Robustness measures for quantifying nonlocality. *New J. Phys.*, 27:053001, 2025. doi:10.1088/1367-2630/adcd5a; arXiv:2311.07077.

[17] IQOQI Vienna. The power of CGLMP inequalities. Open Quantum Problems, Problem 27, 2026. https://oqp.iqoqi.oeaw.ac.at/the-power-of-cglmp-inequalities/, accessed 8 September 2026.

[18] J.-D. Bancal, N. Gisin, and S. Pironio. Looking for symmetric Bell inequalities. *J. Phys. A: Math. Theor.*, 43:385303, 2010. doi:10.1088/1751-8113/43/38/385303; arXiv:1004.4146.

[19] A. Fine. Hidden variables, joint probability, and the Bell inequalities. *Phys. Rev. Lett.*, 48:291–295, 1982. doi:10.1103/PhysRevLett.48.291.

[20] D. Collins and N. Gisin. A relevant two qubit Bell inequality inequivalent to the CHSH inequality. *J. Phys. A: Math. Gen.*, 37:1775–1787, 2004. doi:10.1088/0305-4470/37/5/021; arXiv:quant-ph/0306129.

[21] J. Jesus and E. Zambrini Cruzeiro. Tight Bell inequalities from polytope slices. *Phys. Rev. A*, 108:052220, 2023. doi:10.1103/PhysRevA.108.052220; arXiv:2212.03212.

[22] C. Staufenbiel. Bell inequalities from polyhedral sampling. arXiv preprint, 2026. arXiv:2604.22859.

[23] I. Šupić, D. Cavalcanti, and J. Bowles. Device-independent certification of tensor products of quantum states using single-copy self-testing protocols. *Quantum*, 5:418, 2021. doi:10.22331/q-2021-03-23-418; arXiv:1909.12759.

[24] A. Grudka, K. Horodecki, M. Horodecki, P. Horodecki, R. Horodecki, P. Joshi, W. Kłobus, and A. Wójcik. Quantifying contextuality. *Phys. Rev. Lett.*, 112:120401, 2014. doi:10.1103/PhysRevLett.112.120401; arXiv:1209.3745.

[25] N. Gisin, A. A. Méthot, and V. Scarani. Pseudo-telepathy: input cardinality and Bell-type inequalities. *Int. J. Quantum Inf.*, 5:525–534, 2007. doi:10.1142/S021974990700289X; arXiv:quant-ph/0610175.

[26] P. Joshi, K. Horodecki, M. Horodecki, P. Horodecki, R. Horodecki, B. Li, S. J. Szarek, and T. Szarek. Bound on Bell inequalities by fraction of determinism and reverse triangle inequality. *Phys. Rev. A*, 92:032329, 2015. doi:10.1103/PhysRevA.92.032329; arXiv:1502.03088.

[27] O. Heinävaara. Convolution comparison measures. arXiv preprint, 2026. arXiv:2602.10373.

[28] K. Fukuda and A. Prodon. Double description method revisited. In M. Deza, R. Euler, and I. Manoussakis, editors, *Combinatorics and Computer Science*, volume 1120 of *Lecture Notes in Computer Science*, pages 91–111, Berlin, Heidelberg, 1996. Springer. doi:10.1007/3-540-61576-8_77.